Life v2 — Design pack

Requirements reconciliation · PRD · System design · Diagrams · Design system · Development plan

Generated 2026-10-09 from `docs/` on branch `v2`.

Contents

1. Requirements reconciliation
2. Product requirements (PRD)
3. System design
4. Design system v2
5. Development plan
6. v2 Phase 0 — Docs and design system
7. v2 Phase 1 — Foundation v2
8. v2 Phase 2 — Today
9. v2 Phase 3 — Categories and items
10. v2 Phase 4 — Spiritual: prayers and anchors
11. v2 Phase 5 — Plans: content, university, cadence
12. v2 Phase 6 — Reminders
13. v2 Phase 7 — History and analytics
14. v2 Phase 8 — Motivation: points, streaks, badges
15. v2 Phase 9 — Social: friends, groups, leaderboards, sharing
16. v2 Phase 10 — Platform challenges
17. v2 Phase 11 — Native wrap (Capacitor)
18. v2 Phase 12 — Public template library
19. Appendix — all diagrams (landscape)

Life v2 — Requirements reconciliation

Captured **2026-10-07**. The owner's new direction is compared against everything documented or built before it. The final, numbered requirement set is assigned to phases.

Precedence: where they disagree, this file supersedes CLIENT_REQUIREMENTS.md (original CRD and Addendum A). Addendum B in that file points here.

Related: PRD · System design · Development plan · Design system v2

---

1. Current requirements (before 2026-10-07)

1.1 Built (phases 01–05)

Area	What exists today	v2 verdict
Platform	npm monorepo: Next.js 16 PWA (Serwist), Fastify API, Drizzle + Postgres (Supabase), sql.js offline store, pull/push sync, 185 tests	**Keep**
Data model	`habit_definitions` + `habit_logs`; one log per habit per day; JSON payloads	**Replace** with Category → Item → Occurrence (§6). History is migrated.
Habits	9 seeded system habits auto-added for the one built-in Owner	**Replace.** They become optional starter templates.
Home	Big ring, 7-day bars, "Up next", numbered category sections with 2-column tiles	**Replace** with To Do / In Progress / Done
Prayer	Fixed 5 prayers; Prayed / Missed / Undo; any order; month heatmap and report	**Keep** the any-order logic and stats. The fixed list and screen are replaced.
Quran	Mulk nightly, Kahf on Fridays	**Becomes** ordinary user items with prayer anchors
Health	Steps (manual, quick add, goal), water (glasses), exercise (type and duration), sleep (bed/wake)	**Becomes** ordinary items with quantity / duration tracking. History is migrated.
Visual design	Space Grotesk + Space Mono; 4 palettes from award sites; tiles; no side bars	**Replace** with the reference-image design (DESIGN_SYSTEM_V2)

1.2 Documented but not built

Source	Requirement	v2 verdict
06A / ACC-01…03	Multi-user accounts, data isolation, per-user offline cache	**Keep.** This is Phase 1 now.
06B / CUS-01…04	Start empty, template gallery, own habits/goals/targets, own templates, Home driven by own data	**Keep, generalised:** categories and items replace "habits"
07	Make bed, no-phone timer, plan tomorrow	**Merged.** The first two become user items; "plan tomorrow" becomes SCH-06 / REM-06.
08	Daily tasks, weekly goals, monthly goals (e.g. savings)	**Merged** into ITM (task / goal types)
09	Content pipeline Idea → To Shoot → Editing → Ready → Scheduled → Uploaded, enforced	**Replaced** by flexible plans (PLN)
10	Calendar day detail	**Merged** into the Today date strip and history

Source	Requirement	v2 verdict
11	Analytics dashboard	**Kept** as Progress page (HIS)
12	Per-category reminder toggles, quick add	**Changed** to per-item reminders with priority escalation (REM). The **+** button covers quick add.
13	Polish and deploy	**Kept**, spread across phases
14 / LIB-01…05	Public template library	**Kept** as v2 Phase 12. Provenance fields stay in the schema.

---

2. New requirements (owner, 2026-10-07)

Paraphrased from the owner's message. Each statement is traced to numbered requirements in §5.

#	Owner statement
S1	"I don't like this current interface and typography, I want simple interfaces." See the 4 reference images in `UI Images/`.
S2	When Home loads it shows today's **To Do, In Progress, Completed** tasks.
S3	Another page shows the sections (categories). I can add, delete, and edit categories such as spiritual, physical, content creation, mental, and health. Users create their own.
S4	Under each category I can add tasks and goals: prayer, Quran, and sadqa; steps and gym (with a gym routine); diet, meals, and water.
S5	For each task or goal I choose **what to track** and **how important** it is (low / medium / high / critical). Higher importance adds more tracking and "make sure it is completed".
S6	Reminders that notify, like alarms.
S7	Divide the day. Add prayers into the day. Daily tasks vary per day and per user. People write them in the morning or the night before.
S8	Badges and rewards for motivation.
S9	Expanding a category shows its details. Clicking a task shows its details and history.
S10	Content collab: add the day, date, and restaurant; shoot day, edit day, post after payment clears; optional reminders.
S11	University: assignments, quizzes, presentations, teacher meetings, FYP tasks and meetings. Varies per user.
S12	Not every task must be completed. Unfinished ones can be left and the day can still be **100%**.
S13	A category with nothing upcoming (no collab, no shoot) is fine. Later: cadence such as "post with a one-day gap" or "fixed shoot / edit / post weekdays".
S14	Spiritual starts with daily prayers, but not fixed: customise them, their placement, and their location. Expanding shows Fajr, Zuhr, Asr, Maghrib, Isha (customisable), with a ring and **one-click** tick.
S15	Prayers can be marked **in any order**. A missed Fajr doesn't block marking Zuhr or Asr.
S16	Quran tilawat or a specific surah with a time, prayer, or day: "Surah Kahf on Friday after (or before) Zuhr". The task is active only on that day or after that prayer.
S17	Physical: steps, water, gym, and other activities. Steps accept a direct value like 10,000, **+/−** buttons of 500 or 1,000, and typed input.
S18	Every task has detailed history. For prayer: total prayed per week and month, which prayer is missed most, which is regular, and where to improve.

#	Owner statement
S19	Add tasks for **tomorrow** any time (not tied to the date changing). Mark non-mandatory tasks **pending** or **shift** them to the next day or another day.
S20	Future: groups. Invite registered friends, or invite people to sign up and join. Group goals and tasks; completers and groups are rewarded.
S21	Leaderboards for individuals and groups, with rewards.
S22	Platform-set weekly and monthly goals. Finishing in time earns points and badges.
S23	Later: share a streak or achievement (e.g. a prayer streak) with friends in the app.

---

3. Conflicts and resolutions

New direction wins unless stated. Each resolution has a DECISION.md entry dated 2026-10-07.

#	Current	New	Resolution
C1	Space Grotesk + Space Mono, award-site palettes, tile Home; memory rule "colors from `docs/design-references`"	Simple UI from `UI Images/` (S1)	**Replace.** Tokens are sampled from the 4 reference images. Font is **Poppins**, matching ref 3. The rule "colors come from real sources, never invented" is kept; the source is now `UI Images/`.
C2	Home = ring, Up Next, numbered category tiles	Today = To Do / In Progress / Done (S2)	**Replace.** Day progress becomes a ring in the header. Status tiles plus three lists follow ref 2 / ref 3.
C3	Nav: Home · Goals · Calendar · Progress · More	Home + categories page + history (S2, S3, S9)	Nav becomes **Today · Categories · + · Progress · Profile**. Calendar becomes the date strip on Today and the item history.
C4	Fixed CRD items: Mulk nightly, Kahf Friday, 10k steps, 8 glasses, 6:00 wake, make bed, no-phone hour	Everything user-defined (S3, S4, S14)	They become **starter templates**, never auto-added (also CUS-01).
C5	`habit_logs` unique (habit, date); habits only	One-off tasks, goals, plans, future dates, moves, sub-items (S4, S10, S19)	New model: **Category → Item (tree) → Occurrence**. Occurrence is both the dated instance and its log. Existing logs are migrated (§6).
C6	Day completion counts every scheduled habit	Unfinished items may be left; 100% still possible (S12, S13)	**Progress rule:** % = done ÷ required. Skipped, moved, pending, and optional items are not required. Nothing required means 100%.
C7	Content pipeline with an enforced fixed status chain	Custom dated steps, venue, payment gate, cadence (S10, S13)	**Generic plans:** steps are child items with dates and dependencies. Content and University are presets. Cadence produces warnings and suggestions, not hard blocks.
C8	Prayer: fixed 5, Prayed/Missed buttons, own page	Customisable set, order, and placement; ring; one tap; anchors (S14–S16)	Keep the any-order rule, Missed state, and report maths. The list becomes a **group item** with child prayers. Prayer times come from **adhan-js** (location and method in settings) or manual times.
C9	Reminders: per-category toggles (old Phase 12)	Per-item reminders "like alarms" (S5, S6)	Per-item reminders with **priority escalation**, delivered by **Web Push** now. True alarms need the **Capacitor** wrap (Phase 11), because a PWA cannot ring. On iPhone, push requires the app installed to the Home Screen.

#	Current	New	Resolution
C10	Separate phases 07/08/09/10	One item engine	Folded into v2 phases. Old phase docs move to `docs/phases/archive/`.
C11	Phase 14 public template library	Not mentioned	**Kept** as v2 Phase 12, after social. Provenance columns stay from Phase 1.
C12	ACC-02: all data private	Groups, leaderboards, sharing (S20–S23)	**Private by default.** Others see only points, group-goal progress, and things explicitly shared.
C13	Sleep logged as bed/wake on the wake-up date	Not mentioned	Kept as an optional **starter template** (duration tracking; bed/wake in `occurrence.detail`). History is migrated.
C14	Memory rule: "never invent colors"	Owner supplied new references	Still applies. Dark mode has no reference image, so its values are **derived** and marked `(d)`, and the owner reviews them in Phase 0.

3a. Owner answers (2026-10-08)

#	Question	Answer
A-1	Prayer calculation default	**Karachi method + Hanafi Asr**, synced to the user's location. No API key is needed; see PRAYER_TIMES.md. Becomes PRY-10.
A-2	Qada	A prayer not prayed is left unmarked and becomes **missed** at day end. Users can later mark past missed prayers as **qada** to recover them. Becomes PRY-03, PRY-08, PRY-09.
A-3	Leaderboards	Friends and groups first, then a local friends board and a **global** board. Each user chooses whether their achievements and progress are visible. Becomes SOC-04 and SOC-06.
A-4	Sign-in	**Email and password first** (Phase 1). **Google later**, after checking the Google Cloud console and billing constraints; see AUTH_SETUP.md.

4. Glossary

- **Category:** a user-named section (Spiritual, Physical, Content creation, University…).
- **Item:** anything inside a category.
- `task`: one-off, e.g. "Quiz — Data Structures, 12 Oct"
- `habit`: recurring, e.g. "Steps 10,000 daily"
- `goal`: target over a period, e.g. "Save Rs 20,000 this month"
- `plan`: multi-step with dates, e.g. "Collab — Café Aylanto"
- `group`: container, e.g. "Daily prayers"
- **Occurrence:** one dated instance of an item, with its status and value. Statuses: `todo · in_progress · done · skipped · moved · missed`. A task with no date is **pending**.
- **Required:** an occurrence that counts toward the day %. It must not be optional, skipped, moved, or pending.
- **Anchor:** ties an item to a time, a prayer ("after Zuhr"), or a day part ("Morning").

---

5. Final requirements

MUST = needed for the phase to be accepted. **SHOULD** = expected and droppable only with an explicit owner OK.

TOD — Today (home)

ID	Requirement	Pri	Phase	From
TOD-01	Home opens on **today** and shows required and optional occurrences in three groups: **To Do**, **In Progress**, **Done**	MUST	2	S2
TOD-02	Status tiles at the top show counts. Tapping a tile filters the list.	MUST	2	S2
TOD-03	A day ring shows % by the progress rule (FLX-04)	MUST	2	S12
TOD-04	A date strip shows past and future days. Any date can be viewed and planned.	MUST	2	S7, S19
TOD-05	One-tap complete for checks. Inline **+ / −** and typed value for quantities.	MUST	2	S14, S17
TOD-06	Optional grouping by **day part** (e.g. After Fajr / Morning / Evening) instead of status	SHOULD	4	S7
TOD-07	The prayer block shows on Today as one compact row of prayer chips, at the position the user chooses	MUST	4	S14

CAT — Categories

ID	Requirement	Pri	Phase	From
CAT-01	Create, rename, recolor, re-icon, reorder, archive, and delete categories	MUST	3	S3
CAT-02	Starter suggestions (Spiritual, Physical, Health, Mental, Content creation, University, Work). None are forced; a new account starts empty.	MUST	3	S3, CUS-01
CAT-03	The Categories page lists categories with today's ring and counts. A category **expands inline** to show its items, and tapping opens the category detail.	MUST	3	S9
CAT-04	A category with nothing due shows "Nothing today" and is excluded from the day %	MUST	2	S13
CAT-05	Items within a category can be reordered and pinned. A block such as Daily prayers can be placed first.	MUST	3	S14

ITM — Items (tasks, habits, goals, plans, groups)

ID	Requirement	Pri	Phase	From
ITM-01	Item types: task, habit, goal, plan, group	MUST	3	S4
ITM-02	Items can have **children**: a gym routine has exercises, Daily prayers has prayers, a plan has steps	MUST	3	S4, S14
ITM-03	**Priority** is low, medium, high, or critical. It drives sort order, reminder defaults (REM-02), and an end-of-day check for critical items.	MUST	3	S5
ITM-04	**Optional** flag: never counts toward the day % (critical items cannot be optional)	MUST	2	S12

ID	Requirement	Pri	Phase	From
ITM-05	Free notes plus typed extra fields per preset: venue, platform, course, teacher, amount	MUST	3	S10, S11
ITM-06	Edit, archive, and delete. Archived items keep their history.	MUST	3	S3
ITM-07	**Item detail:** today's control, schedule, reminders, and history (HIS-01)	MUST	3	S9

TRK — What is tracked

ID	Requirement	Pri	Phase	From
TRK-01	**Check:** done or not done	MUST	2	S4
TRK-02	**Quantity:** unit and target (10,000 steps, 8 glasses, Rs 20,000). **+ / −** buttons with a configurable step (500 / 1,000 / 1), plus **direct typed entry**	MUST	3	S17
TRK-03	**Duration:** minutes, with an optional target	MUST	3	S4
TRK-04	**Checklist:** sub-steps ticked within one occurrence, such as meals or a short routine	SHOULD	3	S4
TRK-05	**In Progress** means a quantity above 0 but below target, or a task marked *Started*	MUST	2	S2

SCH — When things happen

ID	Requirement	Pri	Phase	From
SCH-01	One-off on a date, with an optional time	MUST	2	S10, S11
SCH-02	Repeating: daily, chosen weekdays, every N days, or monthly. Optional start and end dates.	MUST	3	S4, S13
SCH-03	**Prayer anchor:** before, at, or after a named prayer, with an optional offset. On a recurring item it limits the item to its days and time window (e.g. Kahf: Fridays, after Zuhr until Asr).	MUST	4	S16
SCH-04	Time anchor and user-defined **day parts**	SHOULD	4	S7
SCH-05	**Pending:** a task with no date lives in a Pending list	MUST	2	S19
SCH-06	Items can be planned for **tomorrow or any future date at any time**, e.g. the night before	MUST	2	S7, S19

FLX — Flexible days

ID	Requirement	Pri	Phase	From
FLX-01	**Skip** today's occurrence. It is excluded from the %.	MUST	2	S12
FLX-02	**Move** to tomorrow or a picked date. One-offs move; for a repeating item only that day's instance moves.	MUST	2	S19

ID	Requirement	Pri	Phase	From
FLX-03	**Mark pending:** remove the date and send the item to the Pending list	MUST	2	S19
FLX-04	**Progress rule:** % = done ÷ required. Required excludes optional, skipped, moved, and pending. If nothing is required, the day is 100%.	MUST	2	S12
FLX-05	Categories and plans with nothing upcoming produce no warnings and no penalties	MUST	2	S13
FLX-06	A required occurrence left open at day end is recorded as **missed** for history and streaks only. It never blocks the next day.	MUST	7	S12, S18

PRY — Spiritual and prayer

ID	Requirement	Pri	Phase	From
PRY-01	The prayer set is editable: rename, add (Tahajjud, Jumu'ah, Duha…), remove, and reorder. Default is the five prayers.	MUST	4	S14
PRY-02	The prayer block's placement is editable: its position on Today and in Spiritual, or another category	MUST	4	S14
PRY-03	A **ring** shows the day's prayers, and **one tap** marks each prayer prayed. Long-press (or the visible ⋯) opens Missed, Qada, and Undo. A prayer not prayed today can simply be left **unmarked**.	MUST	4	S14, A-2
PRY-04	**Any order:** no prayer blocks another. Past prayers can be marked later.	MUST	4	S15
PRY-08	**Day end → missed:** a prayer still unmarked when the user's day ends is recorded as **missed**.	MUST	4	A-2
PRY-09	**Qada recovery:** from history, the user can mark any past missed prayer as **prayed (qada)**. It then counts in prayed totals with a qada mark, so old missed prayers can be recovered. On-time streaks count only on-time prayers.	MUST	4	A-2
PRY-10	**Prayer-time defaults** are the Karachi method (University of Islamic Sciences) with Hanafi Asr, computed for the user's location. Each prayer has a per-minute offset to match the local mosque. Setup is in PRAYER_TIMES.md.	MUST	4	A-1
PRY-05	Prayer times come from location, calculation method, and madhab (adhan-js), or are typed in manually. Prayer times work offline.	MUST	4	S16
PRY-06	Quran and surah items use day and prayer anchors, e.g. Kahf on Friday after Zuhr, Mulk after Isha	MUST	4	S16
PRY-07	Prayer insights: totals per week and month, rate per prayer, **most missed**, **most consistent**, and a plain-language suggestion	MUST	4	S18

PLN — Plans (content, university, work)

ID	Requirement	Pri	Phase	From
PLN-01	A plan has **dated steps**. Example: Collab at *Restaurant* has Shoot (date), Edit (date), Payment cleared, and Post (date).	MUST	5	S10
PLN-02	**Dependencies:** a step can wait on another, e.g. Post waits for Payment. A locked step shows why.	MUST	5	S10
PLN-03	Plan fields: venue or place, contact, and notes. Each step can have its own reminder.	MUST	5	S10

ID	Requirement	Pri	Phase	From
PLN-04	**Cadence rules** per category or plan: minimum gap days between steps of a kind, and allowed weekdays for shoot, edit, or post. The date picker warns and suggests; it never blocks.	SHOULD	5	S13
PLN-05	**University preset:** assignment, quiz, presentation, teacher meeting, FYP task or meeting, each with course and due date and time	MUST	5	S11
PLN-06	Plan steps appear on Today on their dates, like any other occurrence	MUST	5	S10

REM — Reminders

ID	Requirement	Pri	Phase	From
REM-01	Per-item reminders: at a time, an offset before due, or relative to a prayer	MUST	6	S6
REM-02	Priority defaults (editable): low = none; medium = one; high = one plus a 30-min follow-up; **critical = repeat every N min until done** (max repeats, respects quiet hours)	MUST	6	S5
REM-03	Notification actions: **Done** and **Snooze**	MUST	6	S6
REM-04	Quiet hours, a global switch, and per-category mute	MUST	6	S6
REM-05	Delivery is Web Push now, with an iOS install prompt. **Exact alarms** come in the native wrap.	MUST	6 / 11	S6
REM-06	Optional "Plan tomorrow" evening nudge and morning summary	SHOULD	6	S7

HIS — History and insights

ID	Requirement	Pri	Phase	From
HIS-01	Item history: month heatmap, 7-day bars, completion rate, current and best streak, and a log list with values	MUST	3 (basic) / 7	S9, S18
HIS-02	Week and month summaries per category and overall	MUST	7	S18
HIS-03	"Needs improvement / doing well" insight per category, generalising PRY-07	SHOULD	7	S18
HIS-04	Past days can be edited (backfill). Late completions do not earn points after 48 h.	MUST	7	S18

REW — Motivation

ID	Requirement	Pri	Phase	From
REW-01	Points per completion, weighted by priority (1 / 2 / 3 / 5), plus a perfect-day bonus. Computed on the server only.	MUST	8	S8
REW-02	Streaks per habit and per day. **Freezes:** earn 1 per 7-day streak, hold at most 2. A broken streak shows a gentle restart, never shaming.	MUST	8	S8

ID	Requirement	Pri	Phase	From
REW-03	Badges at milestones: 3, 7, 30, and 100 days; all prayers for a week; first plan finished; perfect week	MUST	8	S8
REW-04	A small celebration on completing an item, the day, or a badge (respects reduced motion)	SHOULD	8	S8

SOC, CHL, SHR — Social, challenges, sharing

ID	Requirement	Pri	Phase	From
SOC-01	Friends: invite by link or email, accept, remove, block	MUST	9	S20
SOC-02	Groups: create, invite registered friends, or use an invite link that leads to sign-up and then joining	MUST	9	S20
SOC-03	**Group goals and tasks** with a period. Each member logs from their own Today. Completers and the group earn points.	MUST	9	S20
SOC-04	**Leaderboards, in this order:** (1) friends and groups; (2) a local friends leaderboard and a **global** all-users leaderboard. Group vs group uses average points per member. Weekly and monthly, with rewards for the top.	MUST	9	S21, A-3
SOC-05	Privacy: others see only points, group-goal progress, and what was shared	MUST	9	C12
SOC-06	**Visibility setting:** each user chooses who sees their achievements and task progress (Only me / Friends / Everyone). It is honoured by leaderboards, profiles, and shares. Users who choose "Only me" stay off the global board.	MUST	9	A-3
CHL-01	**Platform challenges**, weekly or monthly. Users join and complete them in time to earn points and a badge.	MUST	10	S22
CHL-02	A minimal admin screen to create, schedule, and close challenges	MUST	10	S22
SHR-01	**Share a streak or achievement** with friends in the app; friends can send kudos	MUST	9	S23
SHR-02	Share as an image outside the app	SHOULD	11	S23

UI — Interface

ID	Requirement	Pri	Phase	From
UI-01	Simple interface per `UI Images/` : light lavender-grey background, white rounded cards, pastel status colors, **Poppins**, few labels, no decoration for its own sake	MUST	0 / 2	S1
UI-02	Bottom nav: Today · Categories · **+** · Progress · Profile	MUST	2	S2, S3
UI-03	Mobile-first (430 px design width). Touch targets of at least 44 px. WCAG AA text contrast.	MUST	all	—
UI-04	Dark mode with derived tokens, reviewed by the owner	SHOULD	2	—

Kept from Addendum A

- **ACC-01…03** (accounts, isolation, per-user offline cache): **Phase 1**.
- **CUS-01…04:** merged into CAT/ITM above.
- **LIB-01…05: Phase 12**. Provenance columns exist from Phase 1.

Non-functional

ID	Requirement
NFR-01	**Offline-first.** Every write lands locally first and syncs later. Today renders from local data in under 1 s on a mid-range phone.
NFR-02	**Privacy:** all user data is scoped by `user_id` on every query. Sync rejects rows the caller doesn't own (fixes BUG-002).
NFR-03	**Time:** dates are the user's local `YYYY-MM-DD`; instants are UTC. Each user has an IANA timezone.
NFR-04	**Integrity:** points, badges, and leaderboards are computed only on the server, idempotently (unique source per ledger row).
NFR-05	**Storage:** the local DB persists to IndexedDB, which removes the 5 MB localStorage cap.
NFR-06	**Accessibility:** AA contrast, focus states, labels on icon buttons, and reduced-motion support.

---

6. Migration of existing data (Phase 1)

Old (`habit_definitions` slug → logs)	New
category values (`faith`, `health`, `personal`…)	One **category** per distinct value per user (Faith → "Spiritual")
`prayer` + payload `{prayers: {fajr: completed|missed|pending …}}`	Group item "Daily prayers" with 5 child items. Each day → 5 occurrences (`completed` →done, `missed` →missed, `pending` →no row).
`quran-mulk`, `quran-kahf`	Habits; Kahf gets weekdays [Fri] and an after-Zuhr anchor; Mulk gets an after-Isha anchor
`steps`, `water` (counter payload)	Quantity habits (step 500 / 1); value copied; target from `config.goal`
`exercise`	Duration habit; `detail.type` keeps the exercise type
`sleep` (`bedAt`, `wakeAt`)	Duration habit; value = minutes; `detail` keeps `bedAt` and `wakeAt`
`make-bed`, `no-phone-hour`, custom boolean habits	Check habits

The migration is a script plus a test with fixture logs. Old tables are kept read-only for one release, then dropped.

---

7. Requirements by phase

Phase	Requirements
0 Docs and design system	UI-01 (tokens and UI kit)
1 Foundation v2	ACC-01…03, NFR-01…05, §6 migration
2 Today	TOD-01…05, CAT-04, ITM-04, TRK-01, TRK-05, SCH-01, SCH-05, SCH-06, FLX-01…05, UI-02, UI-04
3 Categories and items	CAT-01…03, CAT-05, ITM-01…03, ITM-05…07, TRK-02…04, SCH-02, HIS-01 (basic)
4 Spiritual	PRY-01…10, SCH-03, SCH-04, TOD-06, TOD-07
5 Plans	PLN-01…06
6 Reminders	REM-01…06 (REM-05 web part)
7 History and analytics	HIS-01…04, FLX-06
8 Motivation	REW-01…04
9 Social	SOC-01…06, SHR-01
10 Platform challenges	CHL-01…02
11 Native wrap	REM-05 (alarms), SHR-02
12 Template library	LIB-01…05

8. Traceability check

Every owner statement S1–S23 maps to at least one ID:

Statement	IDs
S1	UI-01
S2	TOD-01, TOD-02, TRK-05
S3	CAT-01, CAT-02
S4	ITM-01, ITM-02, TRK-01, TRK-03, TRK-04
S5	ITM-03, REM-02
S6	REM-01, REM-03, REM-04, REM-05
S7	TOD-04, TOD-06, SCH-04, SCH-06
S8	REW-01…04
S9	CAT-03, ITM-07, HIS-01
S10	PLN-01…03, PLN-06
S11	PLN-05
S12	FLX-01, FLX-04, ITM-04
S13	CAT-04, FLX-05, PLN-04
S14	PRY-01…03, CAT-05, TOD-07
S15	PRY-04, PRY-08, PRY-09
S16	SCH-03, PRY-05, PRY-06
S17	TRK-02, TOD-05

Statement	IDs
S18	HIS-01…03, PRY-07
S19	SCH-05, SCH-06, FLX-02, FLX-03
S20	SOC-01…03
S21	SOC-04, SOC-06
S22	CHL-01, CHL-02
S23	SHR-01, SHR-02

Life v2 — Product requirements document

Product	Life: a personal planner and tracker for faith, health, study, and work, with friends later
Version	v2 (supersedes the v1 habit dashboard)
Status	Draft for owner review, 2026-10-07
Owner	abdulazizatGitHub
Detail	Numbered requirements: REQUIREMENTS_V2 · Architecture: SYSTEM_DESIGN · Plan: DEVELOPMENT_PLAN · Visuals: DESIGN_SYSTEM_V2

---

1. Problem

People who balance prayer, health, study, and side work like content creation juggle several apps:

- a habit tracker for streaks
- a to-do app for one-off tasks
- a prayer app
- notes for collab plans

Each one assumes every item must be done every day. A missed Fajr or a day with no shoot feels like failure, so people stop using the tracker.

The current Life app tracks a fixed set of habits on a decorative dashboard. The owner finds it visually heavy, and it can't hold one-off tasks, plans, or user-defined sections.

2. Vision

> Open the app and see exactly what today holds — what's left, what's in progress, what's done — in sections *you* defined. Plan tomorrow tonight. Skip what doesn't fit without losing your 100%. Stay motivated with streaks, badges and, later, friends.

3. Users

Persona	Needs
Student creator (primary; matches the owner)	Five prayers and Quran goals, university deadlines (quiz, assignment, FYP), restaurant collabs with shoot, edit, and post dates, steps and water. Plans at night, misses things sometimes, wants honest but kind progress.
Practising professional	Prayer times by location, gym routine, meal and water tracking, work tasks; reminders that matter and nothing noisy
Accountability friends (R3)	A small group sharing goals ("70k steps this week", "all prayers on time"), leaderboard, kudos

4. Goals and success metrics

Goal	Metric (first 8 weeks after R1)	Target
Today is useful daily	Days the app is opened per active user per week	≥ 5
Planning ahead happens	Share of active days with at least one item created the day before	≥ 40%
Flexibility keeps people going	Users active in week 4 who were active in week 1	≥ 50%
Reminders help, not annoy	Reminded occurrences completed within 60 min	≥ 35%
	Notification opt-out rate	< 15%
Prayer tracking works	Users with the prayer block who log at least 3 prayers per day on average	≥ 60%

Non-goals for v2:

- calendar sync with Google or Apple
- team or work project management
- public social feed
- payments
- AI planning

5. Scope by release

Release	Phases	Contents
R0 Plan	0	This pack; UI kit from the reference images
R1 Personal MVP	1–5	Accounts and new data model; **Today** (To Do / In Progress / Done, skip, move, pending, plan ahead); **Categories and items** (tracking, priority, schedules, routines, history basics); **Spiritual** (custom prayer block, any-order one tap, prayer times, anchored Quran items, prayer insights); **Plans** (content collab, university, cadence)
R2 Habit loop	6–8	Reminders with escalation; History and analytics; Points, streaks with freezes, badges
R3 Social	9–10	Friends, groups, group goals, leaderboards, sharing streaks; platform challenges
R4 Native and library	11–12	Capacitor app with real alarms and an external share image; public template library

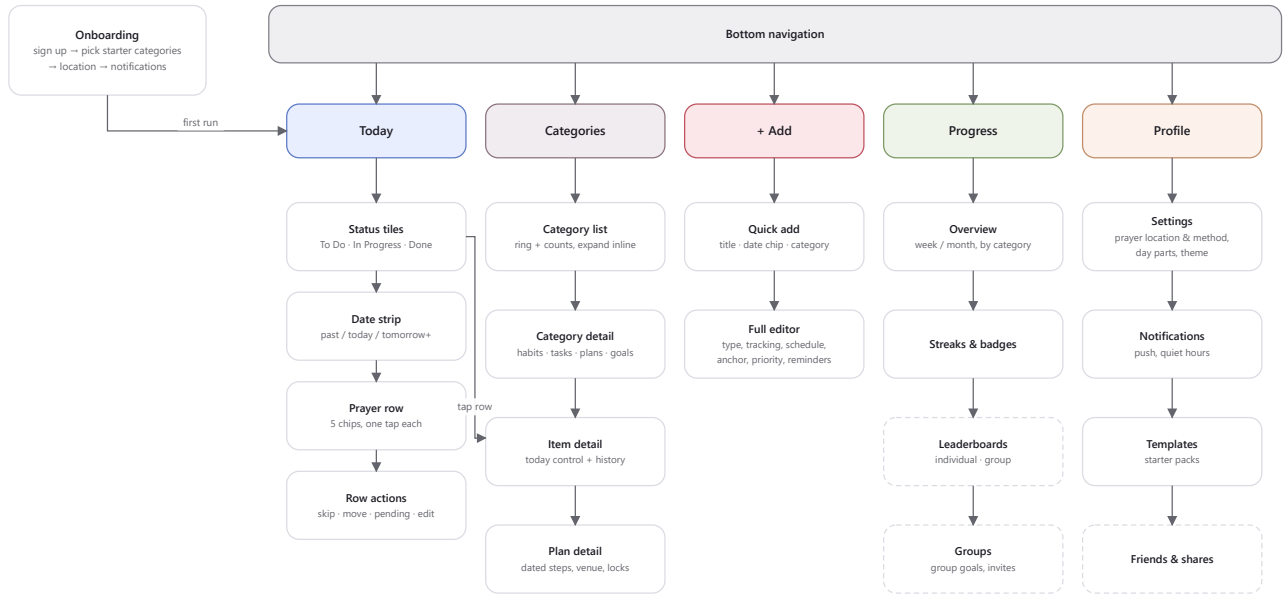
6. Experience

6.1 Navigation

Bottom bar: **Today · Categories · (+) · Progress · Profile**.

Screen map

Bottom navigation: Today · Categories · + · Progress · Profile. Dashed = later phases.



6.2 Key flows

First run

1. Sign up.
2. "What do you want to track?" offers starter categories (Spiritual, Physical, Health, Mental, Content creation, University, Work) plus "Start empty".
3. Optional location for prayer times.
4. Optional notifications, with an install-to-Home-Screen hint on iPhone.
5. Land on Today.

Morning: Today

- Header: date, greeting, and a small day ring.
- Three status tiles (To Do · In Progress · Done) with counts.
- Below them, lists grouped by status, with the prayer row pinned where the user put it.
- Interactions:
- Tap a circle to finish an item.
- Tap **+500** or type **10000** on steps.
- Long-press for Skip, Move to tomorrow, Move to date, Pending, Edit.

Night: plan tomorrow

- Tap the date strip, then tomorrow, then **+**.
- Add "Shoot at Café X, 4 pm, high priority, remind 2 h before".
- It shows on tomorrow's To Do immediately.

Spiritual

- Category detail starts with the "Daily prayers" ring: five chips, each a one-tap toggle in any order.
- Below it, items such as "Surah Kahf · Fridays · after Zuhr", which shows on Fridays only and highlights after Zuhr.
- Tap the prayer block to see insights: "This month 128/150. Most missed: Fajr (9). Strongest: Maghrib (100%). Try a Fajr reminder 10 min after adhan."

Content collab plan

- Use **+**, then Plan, then the *Content collab* preset.
- Fill in restaurant, shoot date, and edit date.
- Post is locked until **Payment cleared** is ticked.
- If the category has the cadence "post weekdays Tue/Fri, 1-day gap", picking Thursday shows "Thursday isn't a post day. Next: Fri 17 Oct."

University

- The *University* preset adds a course name.
- Quick-add Quiz, Assignment, Presentation, Teacher meeting, or FYP meeting, each with a date and time.
- They show on Today on that date and on the Progress page by course.

6.3 Rules users will notice

1. **100% is reachable.** Skipped, moved, pending, and optional items don't count. A day with nothing required is 100%.
2. **Nothing is forced.** No default habits, no fixed prayer order, no mandatory categories.
3. **Windows guide, never lock.** "After Zuhr" items highlight after Zuhr but can be ticked any time.
4. **Kind streaks.** Freezes cover the occasional miss. A broken streak says "Start a new streak", never shames.
5. **Private by default.** Friends see points and shared achievements, never your item list.

7. Functional requirements (summary)

The full table with IDs, priorities, and phases is in REQUIREMENTS_V2 §5.

Area	Must-haves
Today (TOD)	Status buckets, tiles, ring, date strip, one-tap and stepper input, prayer row
Categories (CAT)	User CRUD and reorder, inline expand, "nothing today" without penalty
Items (ITM)	task / habit / goal / plan / group, children, priority, optional, extra fields, detail page
Tracking (TRK)	check, quantity with ± step and typed value, duration, checklist
Scheduling (SCH)	once, repeat rules, prayer anchors, day parts, pending, plan any future date
Flexibility (FLX)	skip, move, pending, progress rule
Prayer (PRY)	custom set, order, and placement; ring; any order; prayer times; Quran anchors; insights
Plans (PLN)	dated steps, dependencies, venue, cadence warnings, university preset
Reminders (REM)	per-item, priority escalation, Done / Snooze, quiet hours; alarms in R4

Area	Must-haves
History (HIS)	heatmap, bars, rates, streaks, summaries, backfill
Motivation (REW)	points, streaks with freezes, badges, celebration
Social (SOC/SHR/CHL)	friends, groups, group goals, leaderboards, shares and kudos, platform challenges

8. Non-functional requirements

- **Offline-first:** works with no network; syncs later. Today renders from the device in under 1 s.
- **Privacy:** data isolated per account; server-side ownership checks; social sees points only.
- **Accessibility:** WCAG AA contrast, 44 px targets, labelled icon buttons, reduced motion.
- **Time correctness:** per-user timezone; local dates; UTC instants; correct across DST.
- **Platforms:** modern Chrome, Edge, Safari, and Firefox. Installable PWA. iOS push only when installed.

9. Design direction

From the owner's 4 reference images (`UI Images/`):

- light lavender-grey canvas, white rounded cards, soft shadows
- pastel status colors (coral To Do, peach In Progress, sage Done)
- Poppins, thin rings with the % inside, slim rounded bars
- a center **+** in the bottom bar

Details and tokens are in DESIGN_SYSTEM_V2.

10. Dependencies

Dependency	Use	Note
Supabase (Postgres, Auth)	Data and accounts	Already in use for Postgres
adhan-js	Prayer times on device and server	MIT, astronomical algorithms (Meeus)
web-push	VAPID push from the API	VAPID keys in env
pg-boss	Scheduled jobs	Uses the same Postgres
Capacitor (R4)	Native wrapper, local notifications	Requires app-store accounts

11. Risks and mitigations

Risk	Mitigation
Reminders on iPhone need install; EU standalone PWAs get no push	Install prompt, in-app reminder list, native wrap in R4
Flexibility could make tracking meaningless	Skips and moves are visible in history ("skipped 3× this week"); points require completion
Too many options overwhelm a "simple" app	Quick add needs only a title; everything else sits behind "More options". Presets fill sensible defaults.

Risk	Mitigation
Social features cause comparison stress	Opt-in groups; per-member average for group ranks; no public global board by default

12. Owner decisions (answered 2026-10-08)

1. **Prayer times:** default is the **Karachi method with Hanafi Asr**, synced to the user's location, with per-prayer minute offsets. See PRAYER_TIMES.md.
2. **Qada:** - An unprayed prayer stays unmarked and becomes **missed** when the day ends. - Users can later mark past missed prayers as **prayed (qada)** to recover them. - Qada counts in prayed totals with a qada mark. It does not extend on-time streaks.
3. **Leaderboards:** - Friends and groups come first, then a local friends board and a **global** board. - A **visibility setting** (Only me / Friends / Everyone) controls who sees a user's achievements and progress.
4. **Sign-in:** - **Email and password** at launch (Phase 1). - **Google** is added later, after the Google Cloud console and billing checks in AUTH_SETUP.md.

Life v2 — System design

Status: proposed, 2026-10-07. **Audience:** developers and agents building v2.

Requirements are in REQUIREMENTS_V2 and the product view is in the PRD. Diagram sources live in `diagrams/src/diagrams.py`. Regenerate them with `python docs/architecture/diagrams/src/diagrams.py`; this needs Python 3 and a local Chrome or Edge, and no network.

---

1. Rebuild strategy

Decision: keep the foundation, replace the domain model and the UI. Work happens on branch `v2`. Nothing is reverted, and the old screens stay in git history.

	What	Why
Keep	npm-workspace monorepo; Next.js App Router; Fastify; Drizzle + Postgres (Supabase); Zod in `@life/shared`; Serwist PWA; Vitest + ESLint setup	Works, is tested, and fits v2. A rewrite would cost weeks for no user value.
Keep, generalise	Sync engine (`apps/web/src/lib/sync/sync-engine.ts`) becomes an entity-agnostic outbox. `local-db.ts` persists to IndexedDB instead of localStorage.	The same pull/push idea, without the habit-only tables and the 5 MB cap
Keep the maths	`computeMonthlyPrayerReport`, `isFriday` (`packages/shared/src/prayer.ts`, `quran.ts`); `summarizeCounter` / `summarizeExercise` / `summarizeSleep` (`health-tracking.ts`); `ProgressRing`, `WeeklyBars` logic	Re-pointed at occurrences; existing tests are kept as regression cases
Replace	`habit_definitions` / `habit_logs` schema → §4 schema plus a migration script; `globals.css` themes and fonts; `today-dashboard.tsx`, `prayer-tracker.tsx`, `health/*` cards, bottom nav, route structure	The owner rejected the UI; the data model can't express tasks, plans, moves, or sub-items
Fix on the way	BUG-002 (sync ownership check); B-001 (`drizzle-kit migrate` on Windows)	Needed before multi-user

Why not start from scratch: auth, sync, offline storage, PWA build quirks (Webpack for Serwist, the sql.js WASM postinstall), the DB connection setup, and the test infrastructure were all solved and documented in DECISION.md. Starting over would re-learn them.

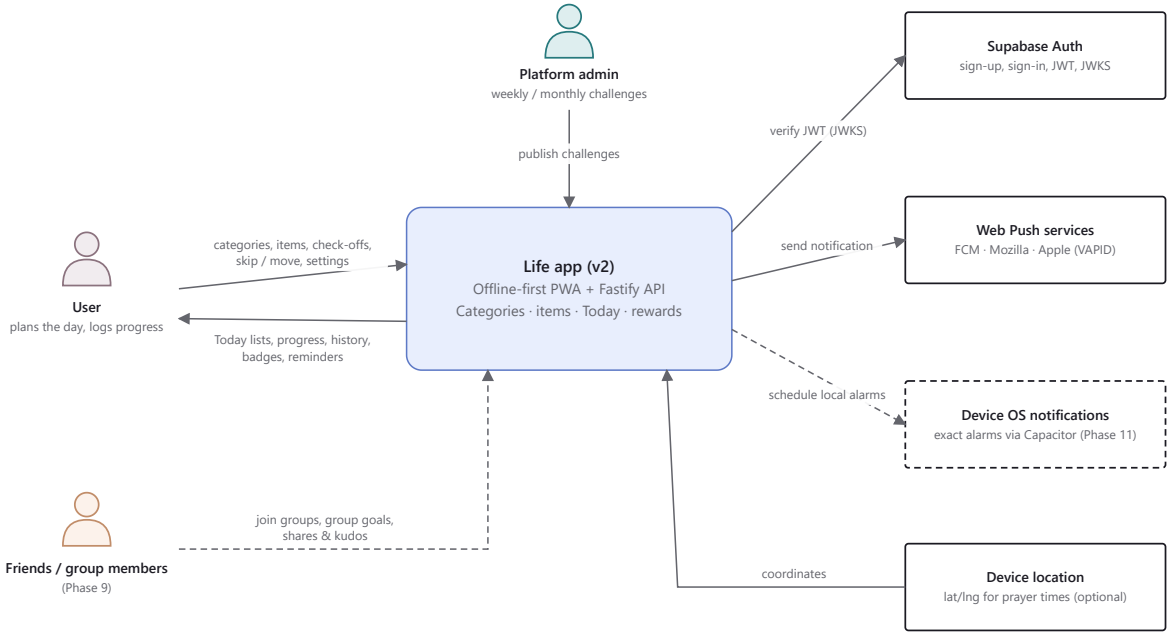
Why not keep going on the current code: the Home, prayer, and health screens are habit-specific, and the owner wants them gone. The one-log-per-day table cannot hold one-off tasks, moved instances, plan steps, or sub-items.

---

2. Context and architecture

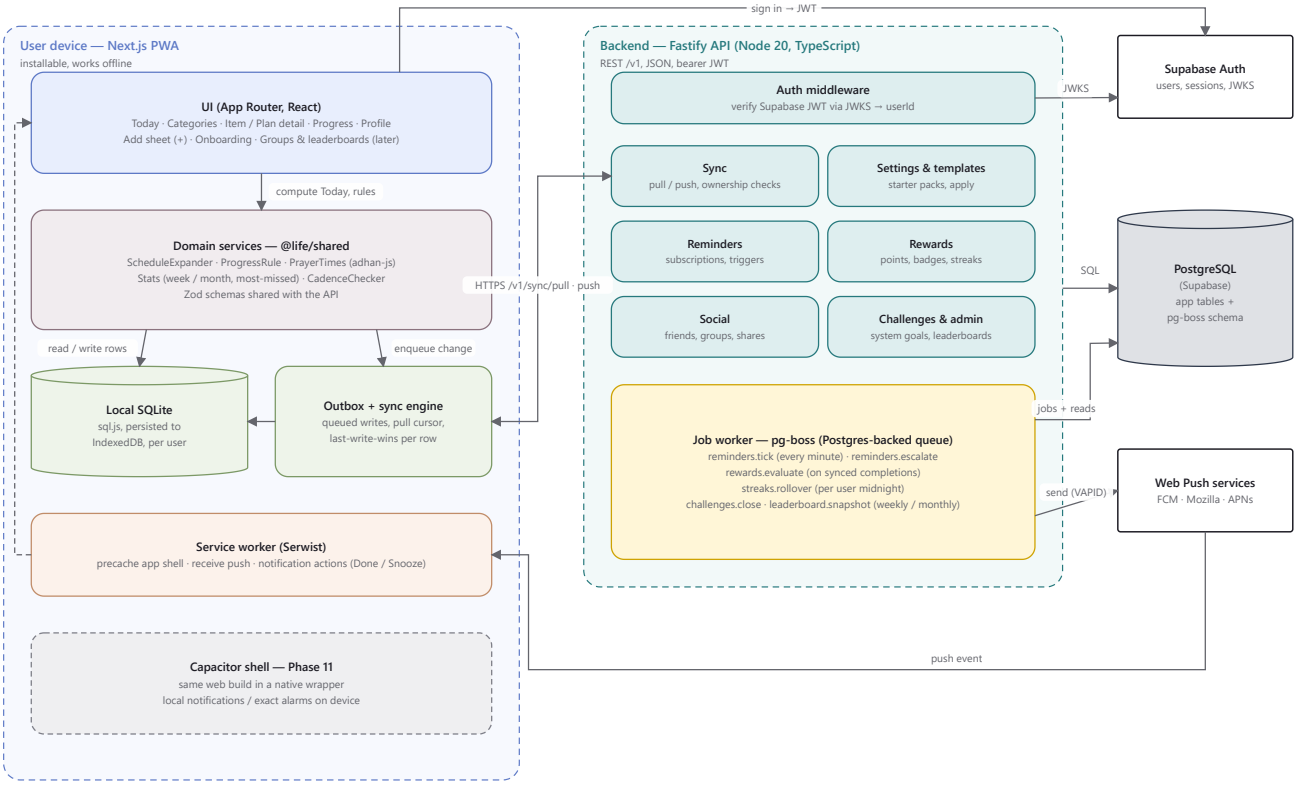
System context

Who and what the Life app talks to. Solid = now (PWA), dashed = later phases.



System architecture

Containers and modules. Postgres is the source of truth; each device keeps a per-user SQLite cache and an outbox.



Container	Tech	Responsibilities
PWA (device)	Next.js 16 App Router, React 19, Tailwind 4, Serwist	UI. Domain services from `@life/shared`. Local SQLite (sql.js → IndexedDB, one DB per user). Outbox. Service worker for offline shell, push, and notification actions.

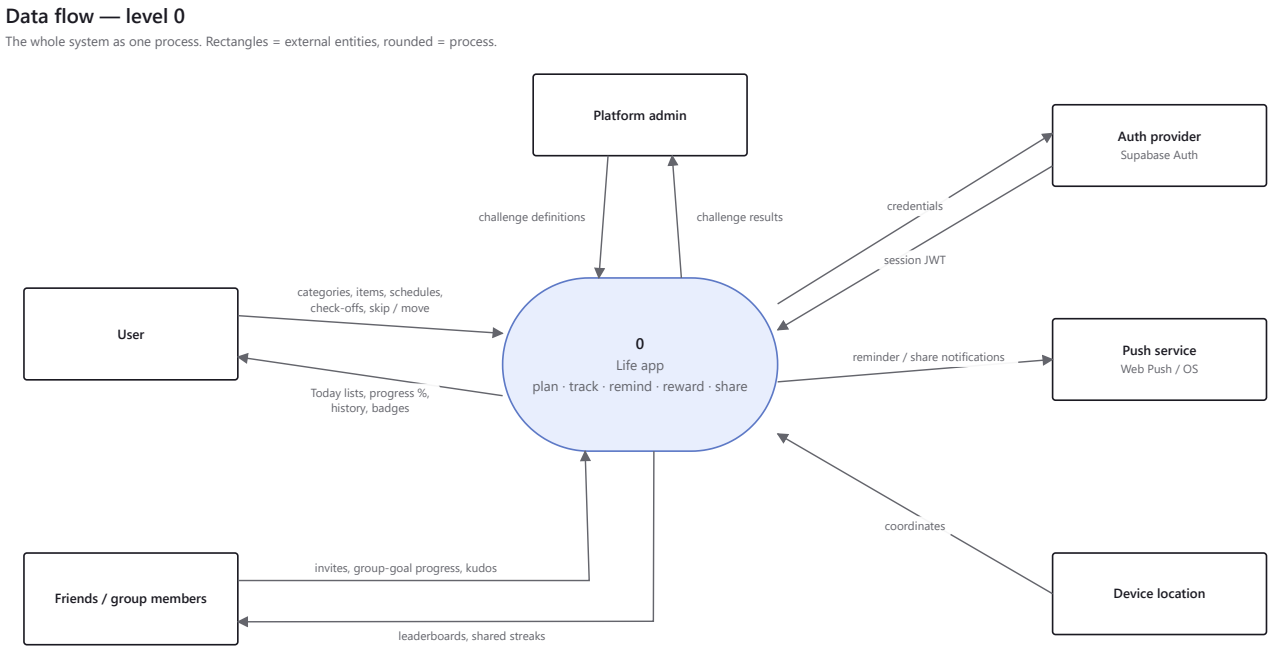
Container	Tech	Responsibilities
API	Fastify on Node 20, Drizzle, Zod	Verifies the Supabase JWT (JWKS, cached) to get `userId`. Runs sync with ownership checks, settings and templates, reminders, rewards read APIs, social, and challenges.
Jobs	pg-boss (queue tables in the same Postgres)	`reminders.tick` every minute, `reminders.escalate`, `rewards.evaluate`, `streaks.rollover` at each user's midnight, `challenges.close`, `leaderboard.snapshot`
Data	Postgres (Supabase)	Source of truth for all tables (§4)
Auth	Supabase Auth	Email and password, password reset, optional Google sign-in. The API never stores passwords.
Push	Web Push (VAPID) via the `web-push` npm package	Chrome / Android / desktop; iOS 16.4+ **only when installed to the Home Screen**. Not available in EU standalone PWAs since iOS 17.4.
Later	Capacitor shell (Phase 11)	Same web build; local notifications and **exact alarms**; share sheet

Choices made here (logged in DECISION.md):

- **Supabase Auth** over self-hosted auth. Postgres is already on Supabase. JWT verification via JWKS keeps Fastify stateless. Accounts map 1:1 to `users.id`.
- **pg-boss** over Redis/BullMQ. There is no new infrastructure, jobs are transactional with app data, and the volume is small.
- **Server-computed rewards.** Clients sync facts (occurrences). Points, badges, streaks, and ranks are derived on the server, so a modified client cannot grant itself points (NFR-04).

---

3. Data flow



Data flow — level 1

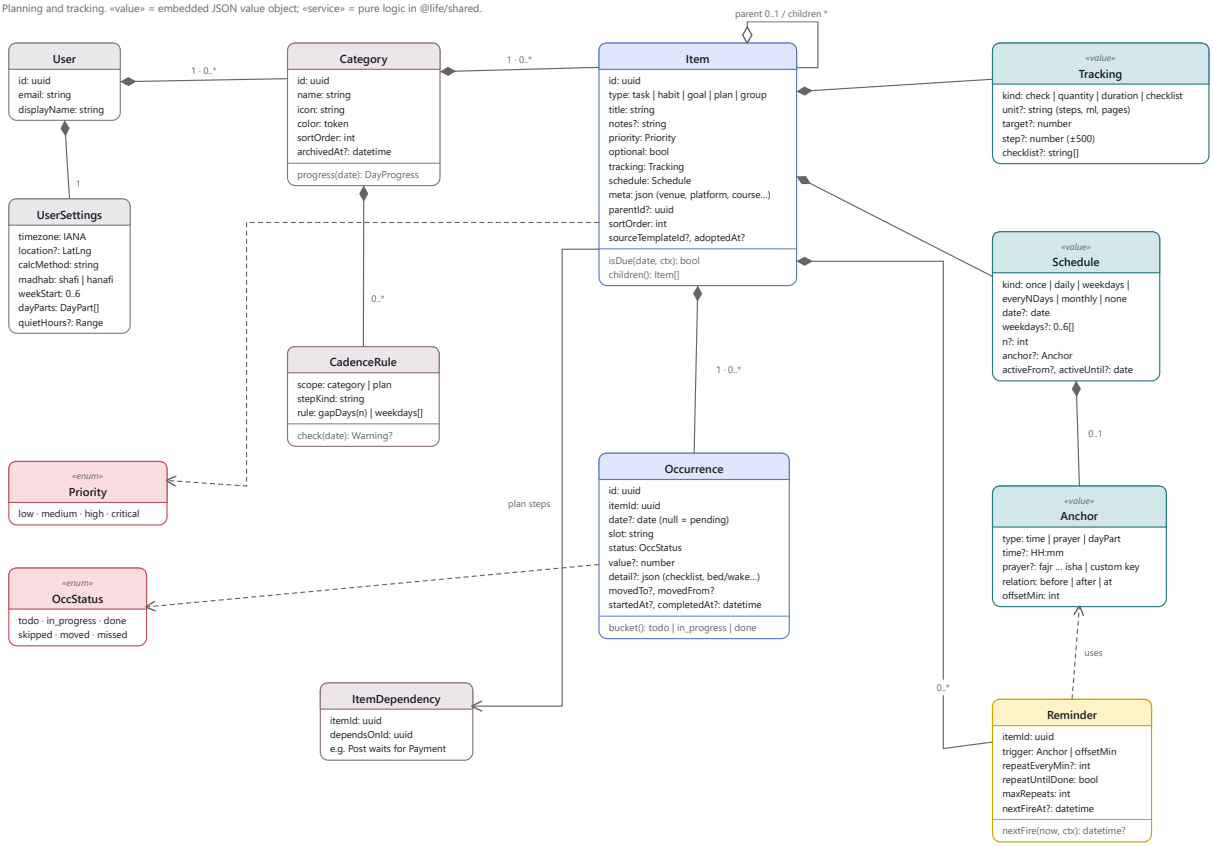
Processes 1–8, data stores D1–D7. Stores are logical: each exists on the server, and D1–D3 + D7 are cached on the device.



4. Data model

Class diagram — core domain

Planning and tracking. «value» = embedded JSON value object; «service» = pure logic in @life/shared.

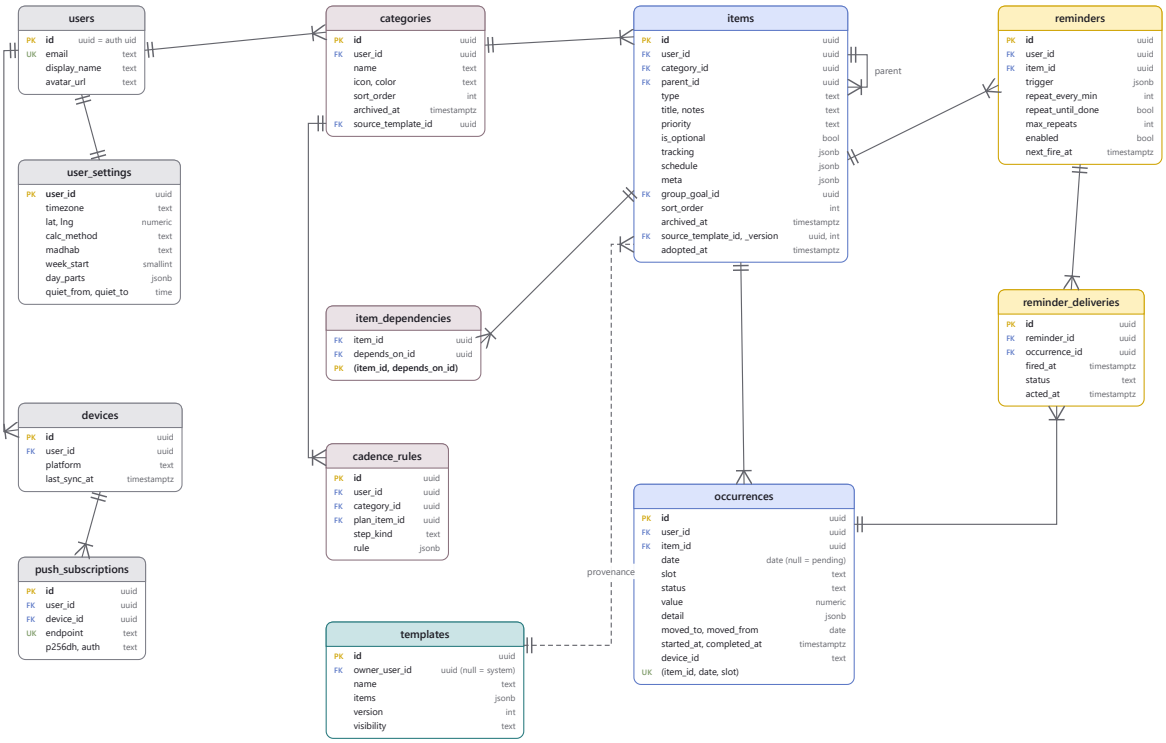


«service» classes operate on the entities above (dependencies omitted for clarity)

«service» ScheduleExpander	«service» ProgressRule	«service» PrayerTimeService	«service» CadenceChecker	«service» ReminderScheduler	«service» SyncEngine
expand(items, date, times)	progress(occurrences)	times(date, settings)	validate(plan, date)	nextFire(reminder, now)	push() · pull(since)

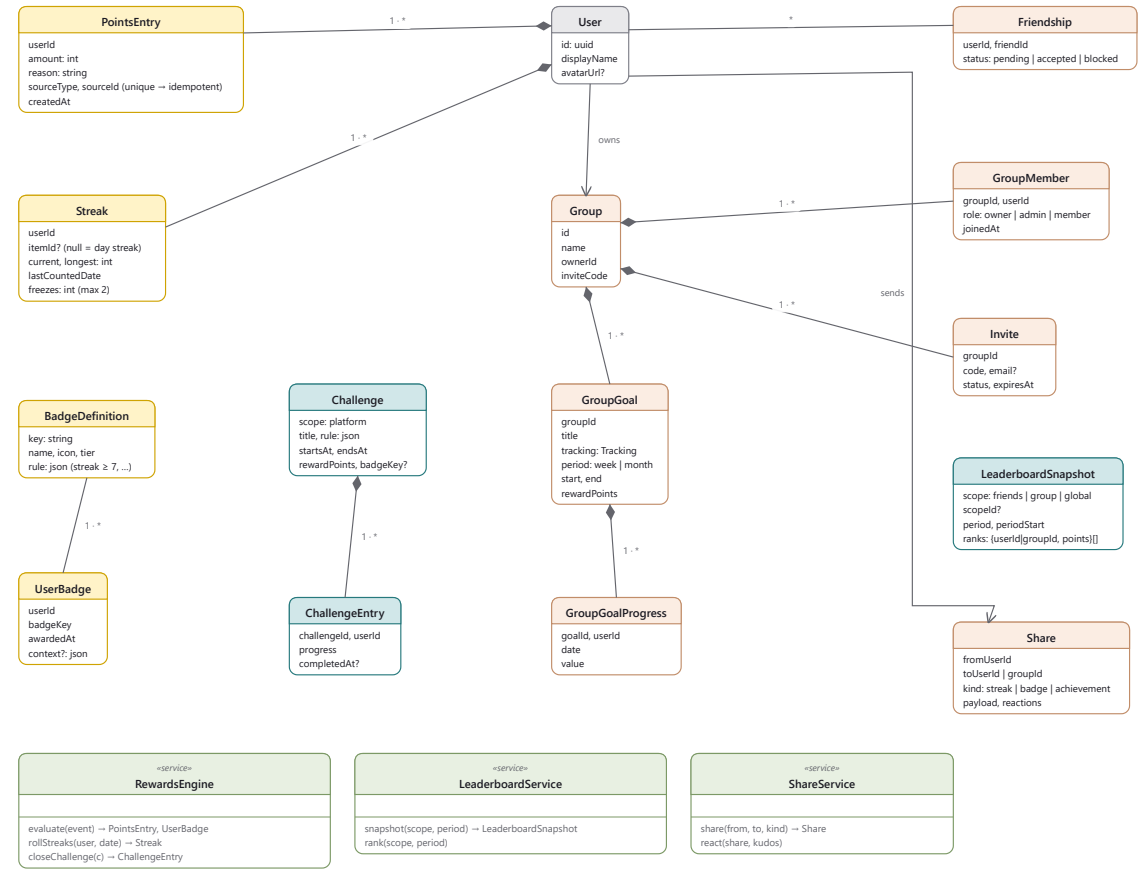
ERD — core tables

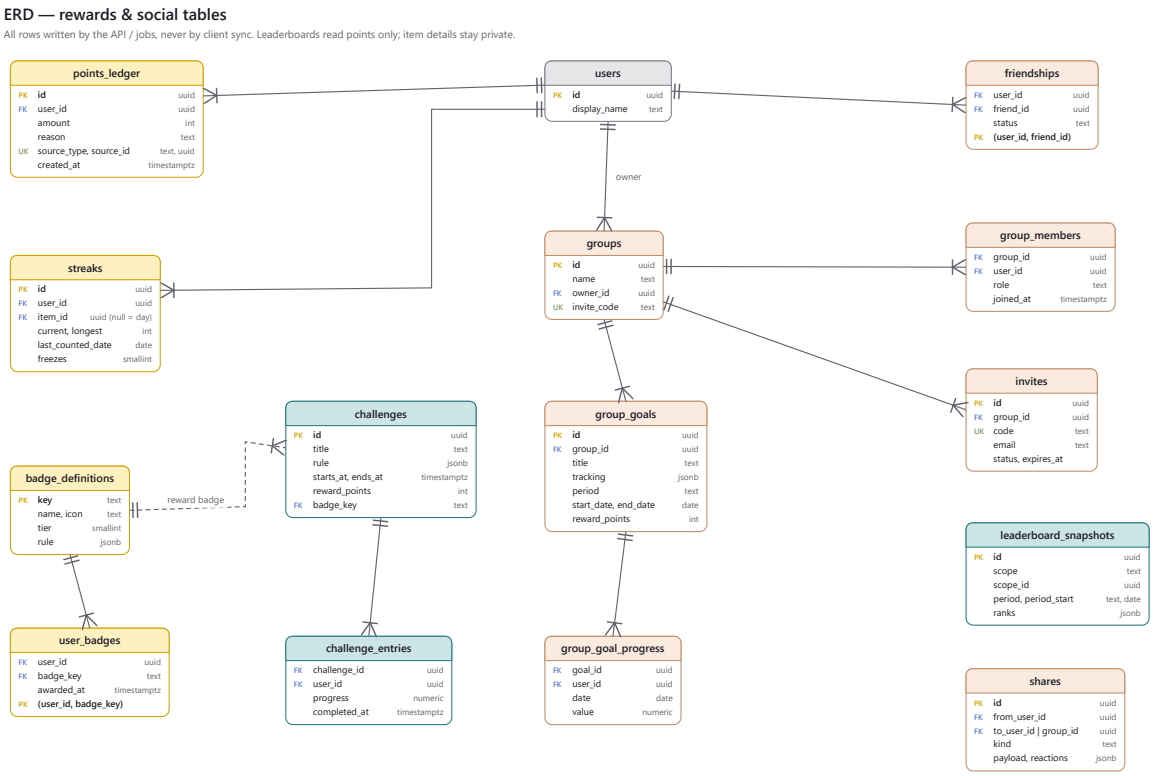
Postgres. Every table also has created_at, updated_at, deleted_at (soft delete, for sync). Local SQLite mirrors the user-owned tables.



Class diagram — rewards & social

Motivation (Phase 8), social (Phase 9) and platform challenges (Phase 10). All written by server-side services only.





4.1 Rules

- Every user-owned table has `user_id`, `created_at`, `updated_at`, and `deleted_at` (soft delete, so deletions sync).
- **Client-syncable entities:** `categories`, `items`, `item_dependencies`, `cadence_rules`, `occurrences`, `reminders`, `user_settings`.
- **Server-only entities:** everything in rewards and social, `reminder_deliveries`, `push_subscriptions` (registered through an endpoint), and `templates`.
- `items.tracking`, `items.schedule`, and `reminders.trigger` are JSONB validated by shared Zod schemas. They are versioned with a `v` field so the shapes can evolve.
- `occurrences` are unique on `(item_id, date, slot)`. `slot` defaults to `""`; it lets an item occur twice a day if ever needed. A pending task has `date = NULL`.

4.2 JSON shapes (Zod, in `@life/shared`)

```
Tracking  = { v:1, kind:'check' }
          | { v:1, kind:'quantity', unit:string, target?:number, step:number }   // step 500 → ±500
          | { v:1, kind:'duration', targetMin?:number }
          | { v:1, kind:'checklist', steps:string[] }

Schedule  = { v:1, kind:'once', date:string|null, time?:string }                 // date null = pending
          | { v:1, kind:'daily'|'weekdays'|'everyNDays'|'monthly',
              weekdays?:number[], n?:number, monthDay?:number,
              anchor?:Anchor, activeFrom?:string, activeUntil?:string }

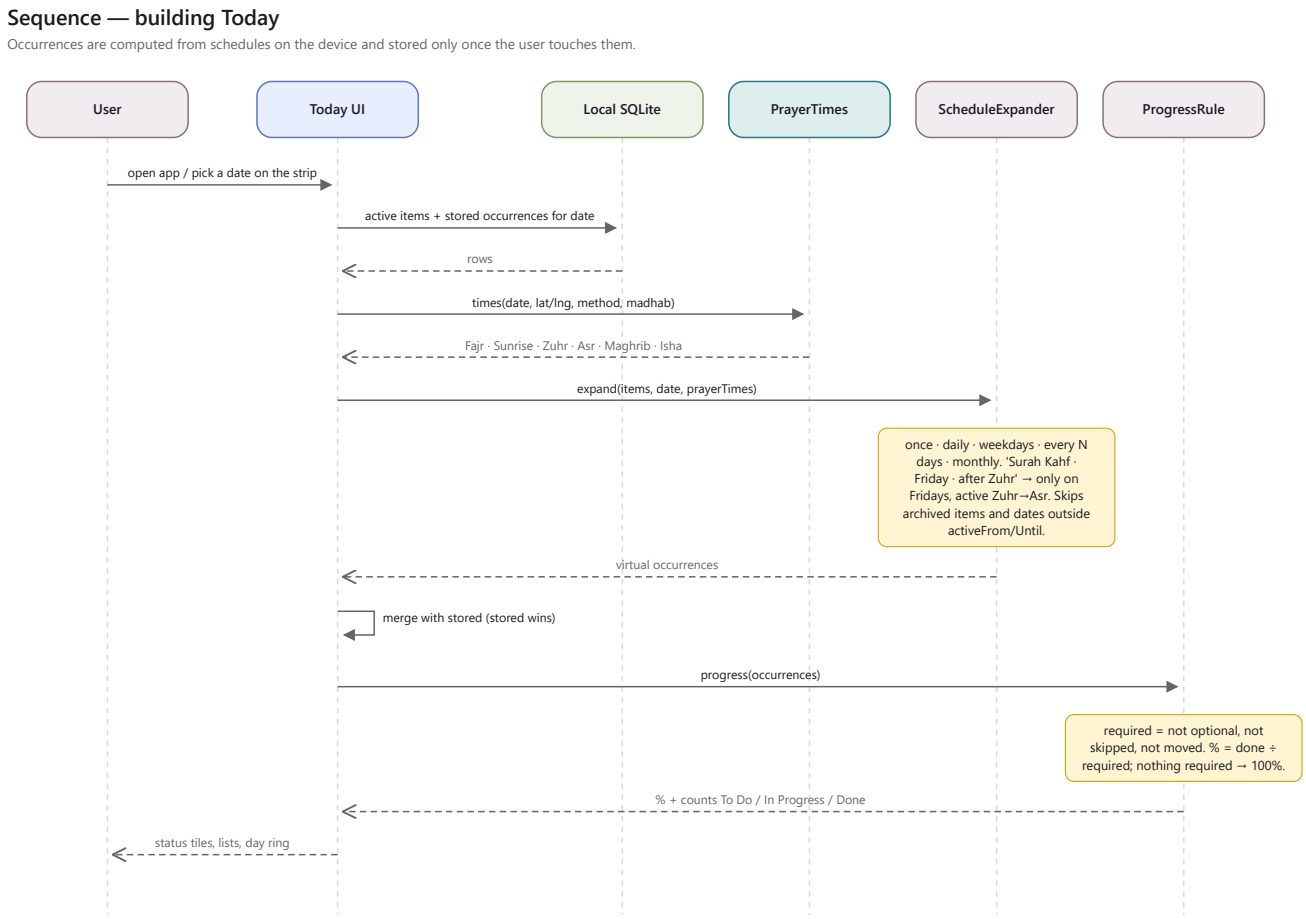
Anchor    = { type:'time', time:'HH:mm' }
          | { type:'prayer', prayer:'fajr'|'sunrise'|'dhuhr'|'asr'|'maghrib'|'isha'|string,
              relation:'before'|'at'|'after', offsetMin:number }
          | { type:'dayPart', key:string }

Trigger   = { type:'at', time:'HH:mm' } | { type:'beforeDue', min:number } | Anchor
```

---

5. Core behaviour

5.1 Building Today



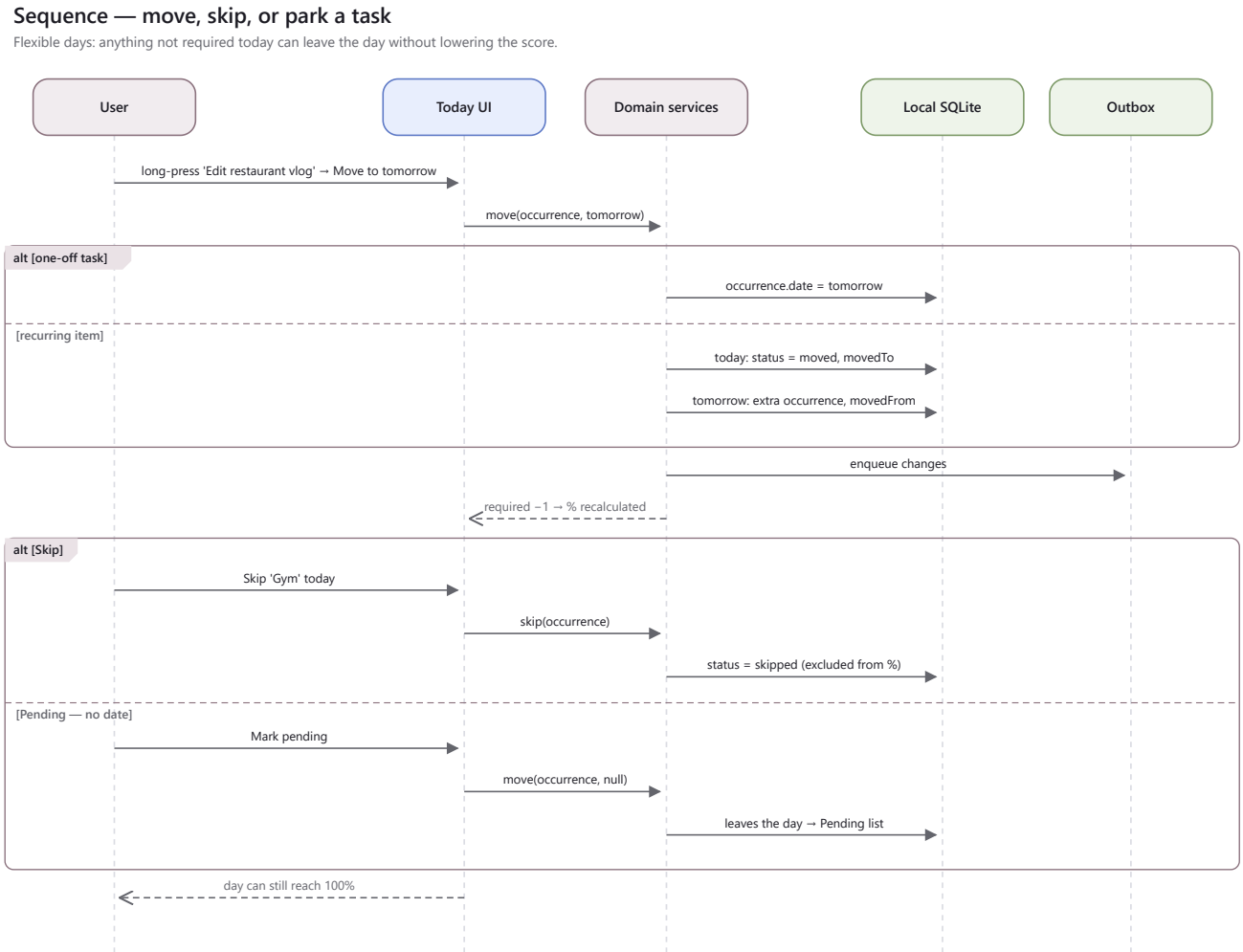
1. Load active items and the stored occurrences for the date from local SQLite.

2. Compute prayer times for the date (adhan-js) from `user_settings`. If no location is set, use manual times.
3. `ScheduleExpander.expand(items, date, times)` yields **virtual occurrences** for items due that day. - Virtual occurrences are stored only when touched (ticked, valued, skipped, or moved). This keeps the table small and makes schedule edits apply retroactively to untouched days.
4. Merge: a stored occurrence wins over its virtual twin. Stored occurrences whose item is no longer due (moved-in, one-offs) are added.
5. Sort into **To Do / In Progress / Done**: - **Done:** `done` - **In Progress:** `in_progress`, or a quantity with `0 < value < target` - **To Do:** everything else that is not skipped or moved - Skipped and moved occurrences show in a collapsed "Not today" row.
6. `ProgressRule`: - `required` = occurrences where `!optional && status ∉ {skipped, moved}`, excluding pending items - `pct = done / required` - `required = 0` → 100%

Anchor windows:

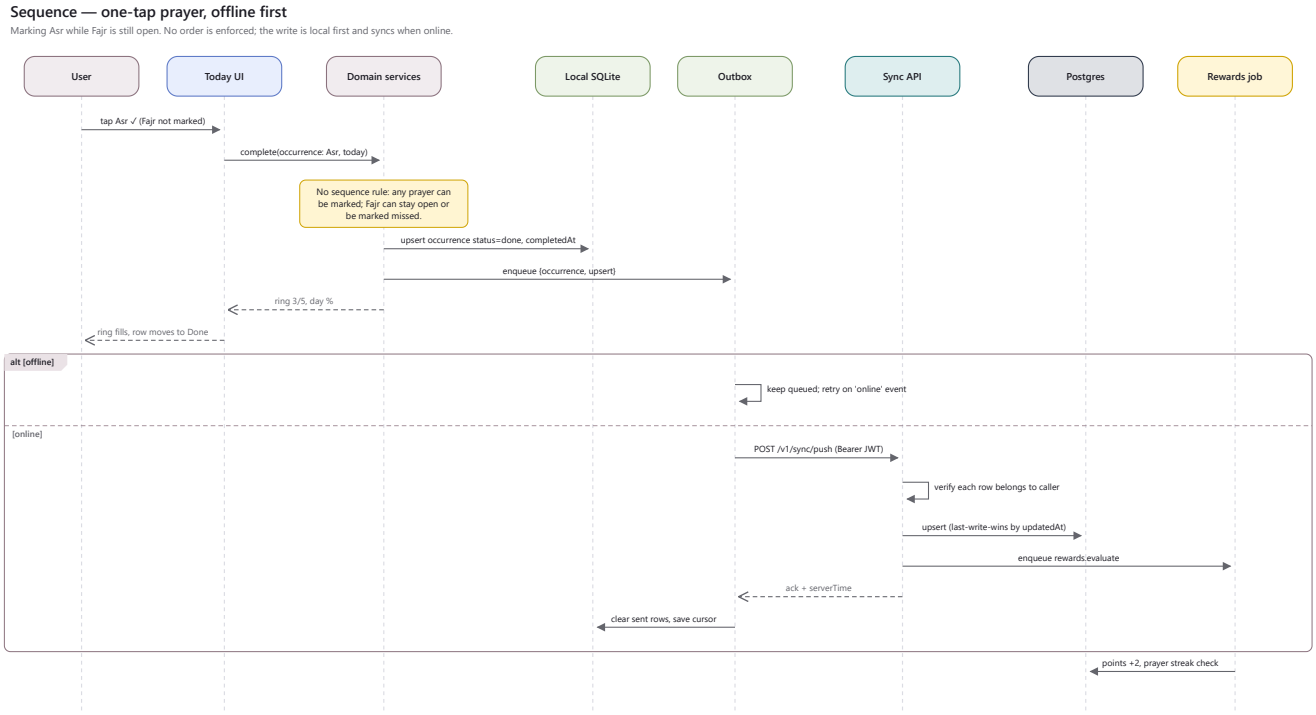
- "after Zuhr" is active from Zuhr until the next prayer time; "before Zuhr" from the previous prayer until Zuhr; "at" ±30 min.
- The window drives ordering and the "now" highlight, and reminders when set.
- A user can still tick an item outside its window; windows guide, they never lock.

5.2 Flexible day operations



Action	One-off task	Repeating item
Skip	`status = skipped`	Same, for that date only
Move to date D	`date = D`	Today: `status = moved, moved_to = D`; plus a new occurrence on D with `moved_from`
Pending	`date = NULL`	Not offered (use Skip)
Missed	Never set by the user for tasks. At day rollover, unresolved required occurrences count as missed in stats only.	Same. Prayers can also be marked Missed explicitly.

5.3 Prayer



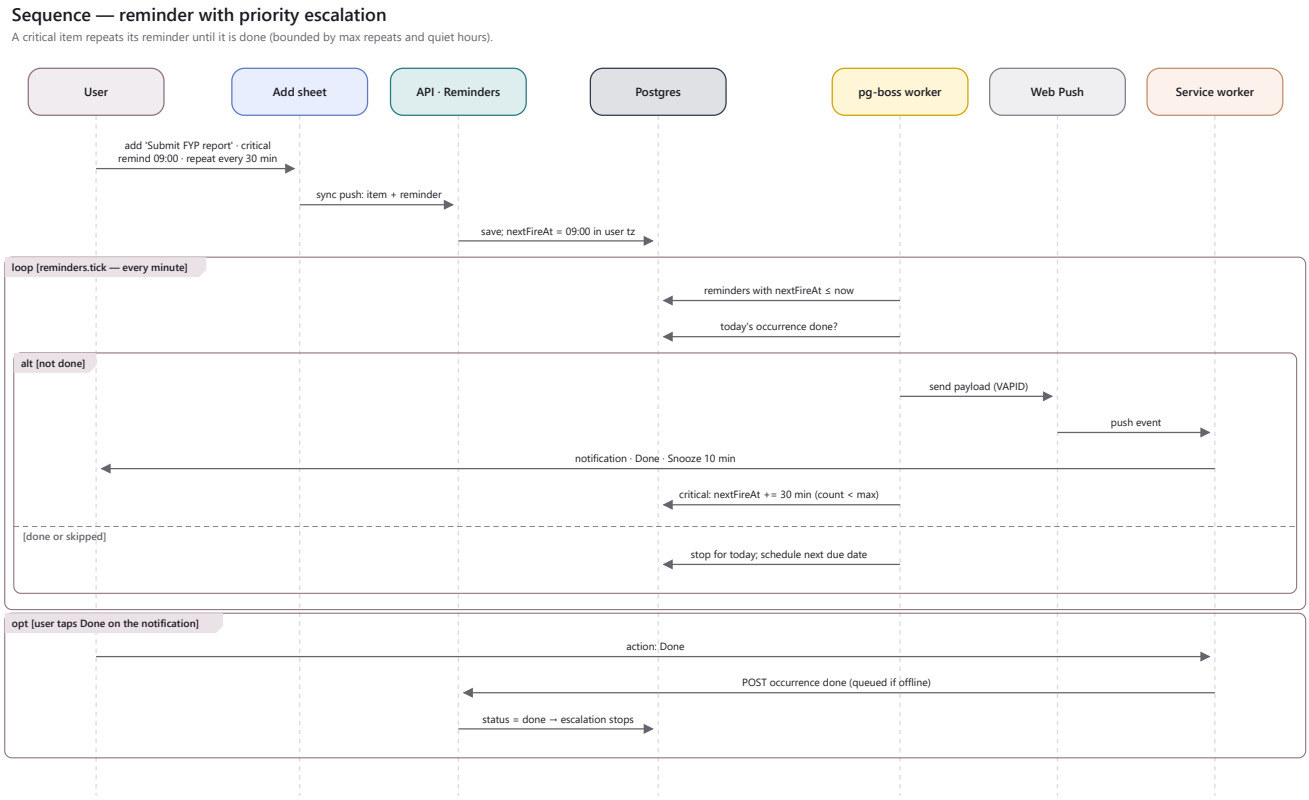
- "Daily prayers" is an item of type `group`. Its children are prayer items with `meta.prayerKey`. Users rename, add, remove, and reorder children (PRY-01) and move the group (PRY-02).
- One tap sets the child's occurrence to `done`. Long-press opens Missed, Qada (done late, flagged), and Undo. Order is never enforced (PRY-04).
- The ring on the group shows the children's statuses.
- `PrayerInsights` computes week and month totals, rate per prayer, most missed, and most consistent. It extends `computeMonthlyPrayerReport`.

5.4 Plans and cadence

- A plan is an item of type `plan`. Its steps are child items, normally one-off tasks with dates.
- `item_dependencies` rows lock a step until the step it depends on is done. Locked steps show a lock and the reason.
- **Presets** are templates applied in one tap; they're data, not code:
- Content collab: Shoot → Edit → Payment cleared → Post; Post depends on Payment
- University: Assignment, Quiz, Presentation, Teacher meeting, FYP task / meeting

- `CadenceChecker` is a pure function. When a step date is picked, it checks the `cadence_rules` of the plan and its category (`gapDays` , allowed `weekdays`). It returns warnings and the next valid dates and never blocks.

5.5 Reminders



Next fire time. `ReminderScheduler.nextFire(reminder, now, ctx)` is shared code; the API uses it to store `next_fire_at` . Prayer-relative triggers use the same adhan-js code on the server, with the user's location.

`reminders.tick` (every minute):

1. Select due reminders with `FOR UPDATE SKIP LOCKED` .
2. Skip any whose occurrence is done or skipped.
3. Respect quiet hours.
4. Send the push to every subscription of the user.
5. Write `reminder_deliveries` .
6. Set the next `next_fire_at` : - repeat (critical), while the delivery count is below `max_repeats` - otherwise the next due date

Priority defaults (REM-02):

Priority	Default reminders
low	none
medium	1
high	1, plus a follow-up after 30 min
critical	every 30 min until done, max 6

Notification actions. In the service worker, **Done** posts the occurrence, or queues it when offline. **Snooze** sets the next fire to now + 10 min.

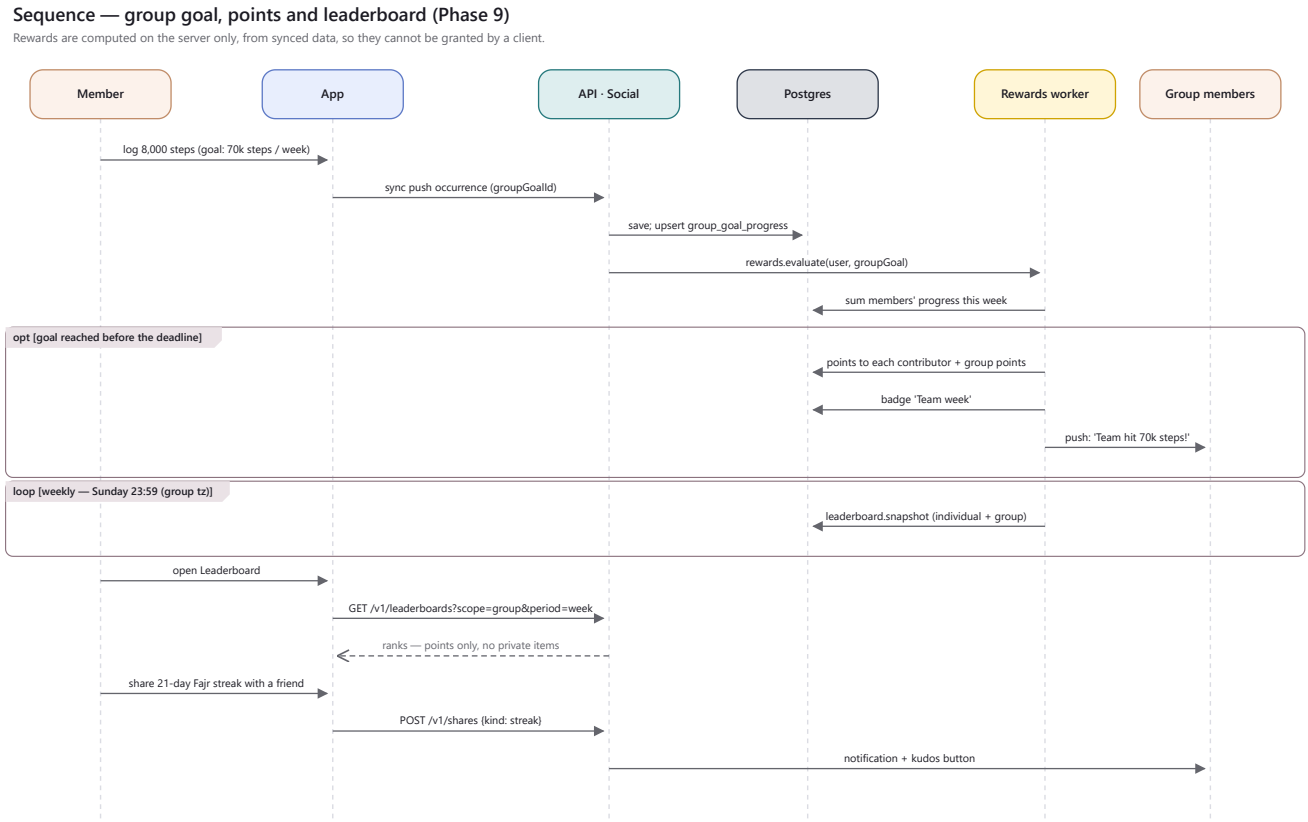
Limits stated to the user:

- iOS needs the app installed to the Home Screen.
- Delivery isn't exact to the second.
- Alarms come with the native wrap.

5.6 Sync

- **Outbox:** each local write appends `{entity, id, op, row, updatedAt}`.
- **Push:** `POST /v1/sync/push` with a batch. The server: 1. validates each row with Zod 2. checks ownership: `row.user_id = caller`, and that every referenced category, item, or parent belongs to the caller (BUG-002) 3. upserts last-write-wins by `updated_at` 4. enqueues `rewards.evaluate` for occurrences that changed to `done`
- **Pull:** `GET /v1/sync/pull?since=cursor` returns changed rows of all syncable entities, including soft deletes, plus `serverTime`.
- **Quantities sync as absolute values,** not deltas. Two devices editing the same day's steps resolve last-write-wins; this is accepted for v2.

5.7 Rewards, social, challenges



Points ledger. Each row has a unique `(source_type, source_id)`, so re-evaluation never double-counts.

Event	Points
Completion, low / medium / high / critical	1 / 2 / 3 / 5

Event	Points
Perfect day	+10
Badge	per definition
Challenge	per challenge
Group goal	per goal, to each contributor

Completions logged more than 48 h after their date earn 0 points (HIS-04).

Streaks:

- per item, and per day (day % = 100)
- `streaks.rollover` runs at each user's local midnight
- freezes: earn 1 per 7-day streak, hold at most 2; a freeze is consumed automatically
- a reset shows "Start a new streak"

Badges. Rules are JSON evaluated by `RewardsEngine.checkBadges`, e.g. `{metric:'streak', item:'prayer-group', gte:7}`.

Leaderboards:

- weekly and monthly snapshots per scope (friends, group, global opt-in)
- group vs group uses **average points per active member**, which is fairer to small groups
- only points and display names are exposed (SOC-05)

Shares:

- a `shares` row plus a push to the recipient
- kudos are stored as `reactions` on the share

---

6. API surface (v1)

All routes require `Authorization: Bearer <Supabase JWT>`, except `/health*`.

Method and path	Purpose	Phase
`GET /health`, `GET /health/db`	Liveness, DB readiness (existing)	—
`GET /v1/me` · `PUT /v1/settings`	Profile, settings (settings also sync)	1
`GET /v1/sync/pull?since=` · `POST /v1/sync/push`	Offline sync of all user-owned entities	1
`GET /v1/templates` · `POST /v1/templates/:id/apply`	Starter packs and presets (creates categories and items with provenance)	3
`POST /v1/push/subscriptions` · `DELETE /v1/push/subscriptions/:id`	Web Push registration	6
`POST /v1/reminders/:id/snooze`	Snooze from a notification	6
`GET /v1/stats/items/:id?period=` · `GET /v1/stats/overview?period=`	Server copy of history stats (the device computes locally too)	7
`GET /v1/rewards`	Points total, level, badges, streaks	8

Method and path	Purpose	Phase
`GET/POST /v1/friends` · `POST /v1/friends/:id/accept` · `DELETE /v1/friends/:id`	Friends	9
`POST /v1/groups` · `GET /v1/groups/:id` · `POST /v1/groups/:id/invites` · `POST /v1/invites/:code/accept`	Groups and invites	9
`POST /v1/groups/:id/goals` · `GET /v1/groups/:id/goals`	Group goals	9
`GET /v1/leaderboards?scope=&period=`	Ranks	9
`POST /v1/shares` · `GET /v1/shares/inbox` · `POST /v1/shares/:id/react`	Sharing and kudos	9
`GET /v1/challenges` · `POST /v1/challenges/:id/join`	Platform challenges	10
`POST/PUT /v1/admin/challenges`	Admin (role claim `admin`)	10

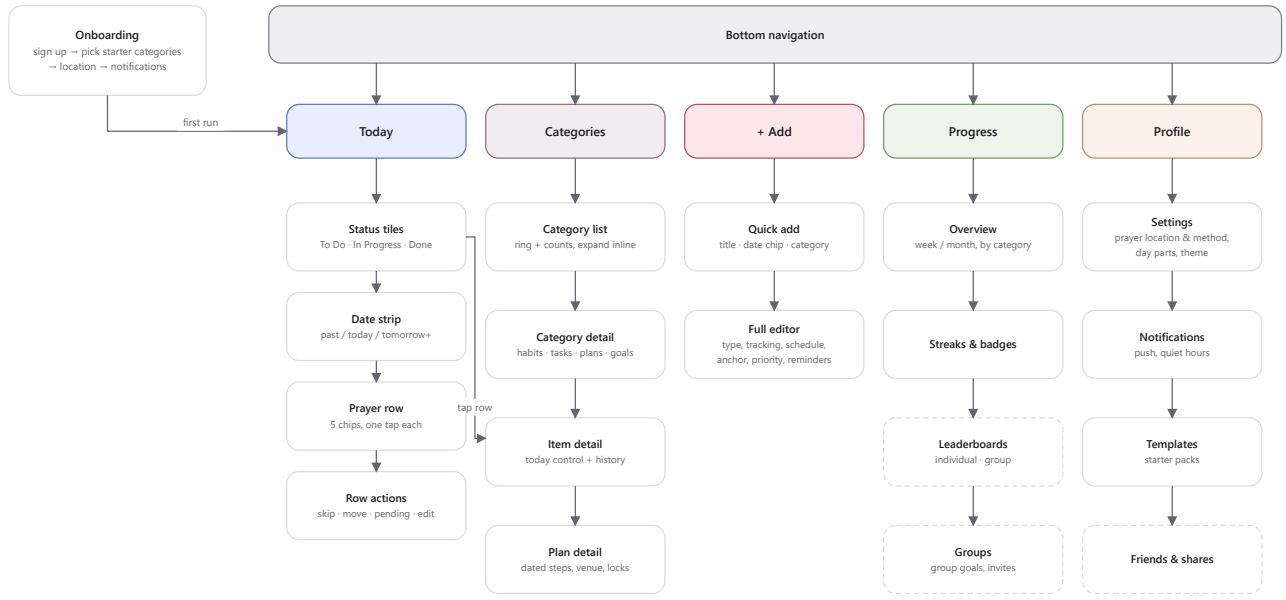
Errors keep the existing shape: `{ error: { code, message } }`.

---

7. Front-end structure

Screen map

Bottom navigation: Today · Categories · + · Progress · Profile. Dashed = later phases.



```
apps/web/src/app/
  (auth)/sign-in, sign-up, reset
  (main)/layout.tsx              bottom nav: Today · Categories · + · Progress · Profile
  (main)/page.tsx                Today
  (main)/categories/page.tsx     list + inline expand
  (main)/categories/[id]/        category detail
  (main)/items/[id]/             item detail + history (prayer group gets its own insight panel)
  (main)/plans/[id]/             plan detail (steps, locks, cadence warnings)
  (main)/progress/               overview, streaks & badges, (later) leaderboards, groups
  (main)/profile/                settings, notifications, templates, (later) friends
  onboarding/
components/ui/                   rebuilt kit from DESIGN_SYSTEM_V2 (Card, StatusTile, Ring, Bars,
Stepper, Chip, Sheet…)
lib/domain/                      thin hooks over @life/shared services
lib/db/, lib/sync/               generalised local store + outbox
```

---

8. Security and privacy

- Identity comes only from the verified JWT. Request bodies never carry `userId`.
- Every query is filtered by `user_id`. Sync push verifies ownership of every referenced id; tests cover cross-user attempts.
- Server-only tables are never accepted from sync.
- Social reads expose display name, avatar, and points only. Item titles are never exposed unless shared.
- Local DB per user; signing out wipes it (ACC-03).
- Rate limiting is on auth-adjacent and social endpoints (`@fastify/rate-limit`).
- The admin role comes from a Supabase custom claim and is checked server-side.

9. Testing

Layer	What	Where
Pure domain	`ScheduleExpander` (incl. anchors, DST, Fridays), `ProgressRule`, `CadenceChecker`, `ReminderScheduler`, `PrayerInsights`, `RewardsEngine` rules	`packages/shared` (Vitest)
API	Ownership rejection, LWW, soft delete, idempotent ledger, leaderboard maths	`apps/api` (Vitest + test DB)
Migration	Fixture old logs → new rows, round-trip counts	`apps/api/scripts` test
UI	Today buckets, one-tap prayer, stepper typed and ± input, move / skip	`apps/web` (RTL)
E2E	Golden flows at 430 px, light and dark	Browser check per phase (`browser-automation`)

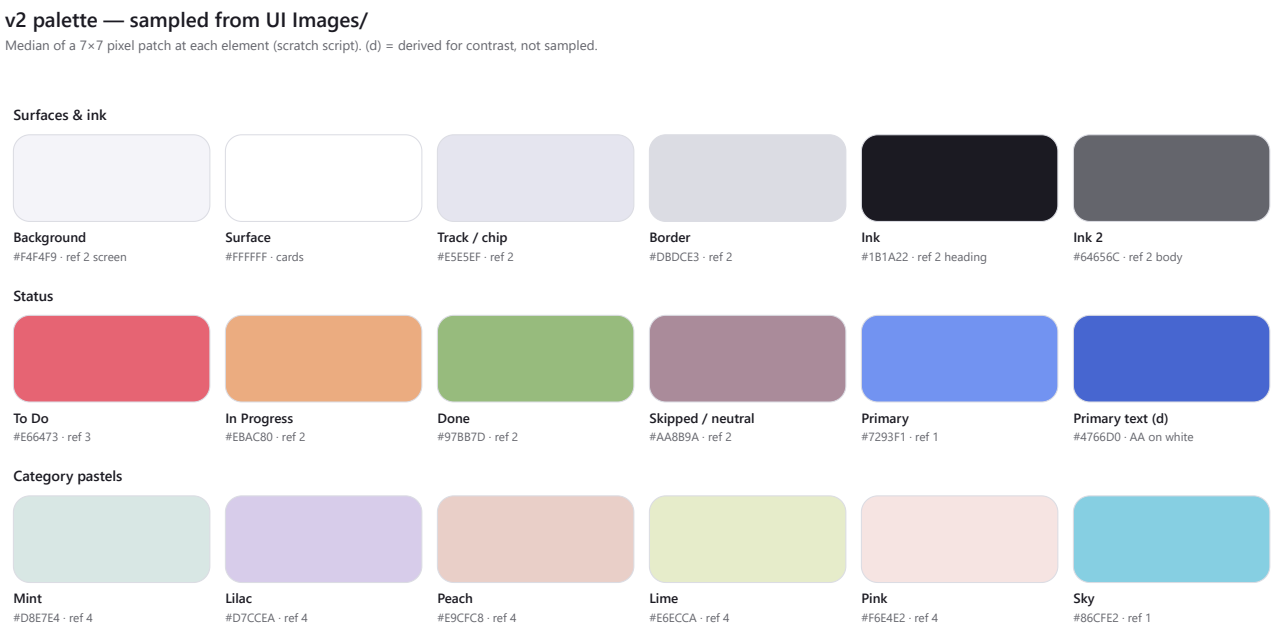
10. Risks

Risk	Mitigation
iOS push only when installed; none in EU standalone	Install prompt, an in-app reminder list as fallback, and the Capacitor phase
Prayer-time accuracy differs by method	Method and madhab in settings; manual override per prayer
Virtual occurrences confuse history when a schedule changes	History reads stored occurrences plus schedule versions. Missed days are computed with the schedule as of that date (`items.schedule` changes are logged in `meta.scheduleHistory`).
Leaderboard cheating via backfill	48 h points window; server-only computation
Scope creep (social before the core is solid)	Releases are gated. R3 starts only after R1–R2 acceptance.

Design system v2: simple, from the owner's references

Status: proposed, 2026-10-07. It replaces the 2026-09-30 visual direction (Space Grotesk, award-site palettes, tiles) per conflict C1 in REQUIREMENTS_V2.

Source of truth: the four images in `UI Images/` at the repo root. Every color below was **sampled** from those images: a scratch script took the median of a 7×7 pixel patch at each UI element. Values marked **(d)** are **derived**, either because no reference covers them (dark mode) or because a sampled value failed WCAG AA for text.



1. What the references have in common

Ref	What it shows	What we take
1 — finance app (3 screens)	Lavender-grey canvas `#E6EAF6`; near-white screens; white cards; one blue accent `#7293F1`; thin donut and slim vertical bars; small pastel icon tiles	Canvas and cards, primary blue, bar and ring style, icon tiles
2 — task manager (Today / Projects / Report)	"Today" heading with a date; **2×2 colored status tiles** (Ongoing / In Process / Complete / Cancel); pill segmented tabs (All / Ongoing / Completed); list rows with checkbox and progress line; project cards with a ring %; bottom bar with a **center +**	Status tiles, segment tabs, list rows, ring-in-card, center + nav, muted status palette
3 — "My Tasks"	**To Do / In Progress / Done** rows with round colored icons and counts; bold geometric sans (**Poppins**); rounded project cards with a ring %	Exact Home structure, typeface, round status icons
4 — pastel habit tracker	Soft mint, lilac, peach, lime, and pink tiles and rows; lots of white space; thin type	Category pastel set, calm density

The rules that follow: one canvas color, white cards, color only where it carries meaning (status, category), one typeface, thin rings and bars, generous spacing, and a single obvious action per row.

2. Tokens

Light (sampled)

Token	Value	Source	Use
`--bg`	`#F4F4F9`	ref 2 screen	App background
`--surface`	`#FFFFFF`	refs 1–3 cards	Cards, sheets, nav
`--track`	`#E5E5EF`	ref 2 canvas	Ring and bar tracks, chips, segmented-control background
`--border`	`#DBDCE3`	ref 2	Hairlines, input borders
`--ink`	`#1B1A22`	ref 2 heading	Primary text
`--ink-2`	`#64656C`	ref 2 body	Secondary text (5.29 : 1 on `--bg`)
`--status-todo`	`#E66473`	ref 3 "To Do"	To Do tile and icon
`--status-progress`	`#EBAC80`	ref 2 "In Process"	In Progress
`--status-done`	`#97BB7D`	ref 2 "Complete"	Done
`--status-muted`	`#AA8B9A`	ref 2 "Ongoing" / FAB	Skipped, moved, neutral
`--primary`	`#7293F1`	ref 1 card and button	Rings, bars, selected states, fills with ink text
`--primary-strong` (d)	`#4766D0`	derived from `--primary`	Filled buttons with white text (5.15 : 1), links
`--cat-mint` · `--cat-lilac` · `--cat-peach` · `--cat-lime` · `--cat-pink`	`#D8E7E4` · `#D7CCEA` · `#E9CFC8` · `#E6ECCA` · `#F6E4E2`	ref 4	Category chips and expanded-category backgrounds
`--cat-sky` · `--cat-teal` · `--cat-yellow`	`#86CFE2` · `#2E9499` · `#FDC703`	ref 1 / ref 3	Category icons and accents

Text on status colors is always `--ink`, never white. Contrast is 5.29 : 1 on To Do, 8.83 on In Progress, and 7.98 on Done. White text on those colors fails AA.

Dark (derived — owner to review in Phase 0)

Token	Value	Basis
`--bg`	`#1B1A22`	Sampled ink, reused as canvas
`--surface` (d)	`#25242E`	`--bg` lifted ~6%
`--track` (d)	`#2F2E3A`	
`--border` (d)	`#3A3946`	
`--ink`	`#F4F4F9`	Sampled light bg (15.7 : 1)
`--ink-2` (d)	`#A9AAB4`	7.47 : 1
Status, primary, category accents	Same as light	Dark ink text on fills, as in light
Category pastels (d)	Each pastel at 30% over `--surface`	Keeps hue, avoids glare

3. Type

- **Poppins** via `next/font/google`, weights 400, 500, and 600. Space Grotesk and Space Mono are removed.
- No uppercase micro-labels above headings. This is the owner's standing rule since 2026-09-30.

Style	Size / line	Weight	Use
Title	24 / 32	600	Screen title ("Today")
Heading	18 / 26	600	Section ("To Do")
Body	15 / 22	400	Row titles
Body strong	15 / 22	500	Item names in detail
Caption	13 / 18	400	Meta: time, category, "after Zuhr"
Number	28 / 32	600	Ring %, stat values

4. Shape, space, elevation

Property	Values
Radius	Cards 20; tiles 20; inputs and buttons 14; chips and segmented controls 999
Spacing	4-pt base. Screen gutter 16. Card padding 16. Row height ≥ 56. Touch targets ≥ 44.
Shadow	One level only: `0 4px 16px rgb(27 26 34 / 6%)` on cards; none in dark mode, where the border is used instead
Rings	6–8 px stroke with round caps on `--track`, % centred in Number style
Bars	6 px wide, fully rounded, on `--track`

5. Components

Component	Spec	Reference
StatusTile	3 across. Status-colored fill, ink count (Number) and label (Caption), round icon top-right. Tap filters.	ref 2 tiles, ref 3 icons
TaskRow	White card row: 28 px round check on the left (empty, half for in-progress, filled for done), title plus caption meta, priority dot (critical = `--status-todo`), chevron. Long-press opens a sheet with Skip / Move / Pending / Edit.	ref 2 list
QuantityRow	TaskRow plus value / target, thin progress line, a **− [value] +** stepper (tap the value to type)	ref 2 progress line
PrayerRow	One card: small ring on the left, then N chips (Fajr…Isha) that each toggle with one tap and show done, missed, or open by fill	ref 3 ring, ref 2 chips
CategoryCard	Pastel icon tile, name, "3 of 5 today", ring on the right. Expands inline to TaskRows.	ref 2 project card, ref 4 pastels
SegmentedTabs	Pill on `--track`; selected = white pill	ref 2
DateStrip	7 days in a row, selected = `--primary` pill with ink text, dot for days with items	ref 1 weekly bars row

Component	Spec	Reference
BottomNav	White bar; 4 icons plus a center **+** (56 px, `--primary-strong` with a white icon)	ref 2
Sheet	Bottom sheet for add, edit, and actions; 20 px top radius	—

Icons stay **lucide-react** (already a dependency), 20 px with a 1.75 stroke.

6. Today layout (430 px)

1. Header: "Today" (Title) and date (Caption) on the left; day ring (48 px) on the right.
2. DateStrip.
3. StatusTiles ×3.
4. PrayerRow, if placed on Today.
5. "To Do" (Heading) with TaskRows, then "In Progress", then "Done" (collapsed after 3 items).
6. "Not today" (skipped and moved), collapsed.
7. BottomNav.

7. Don'ts (carried over and new)

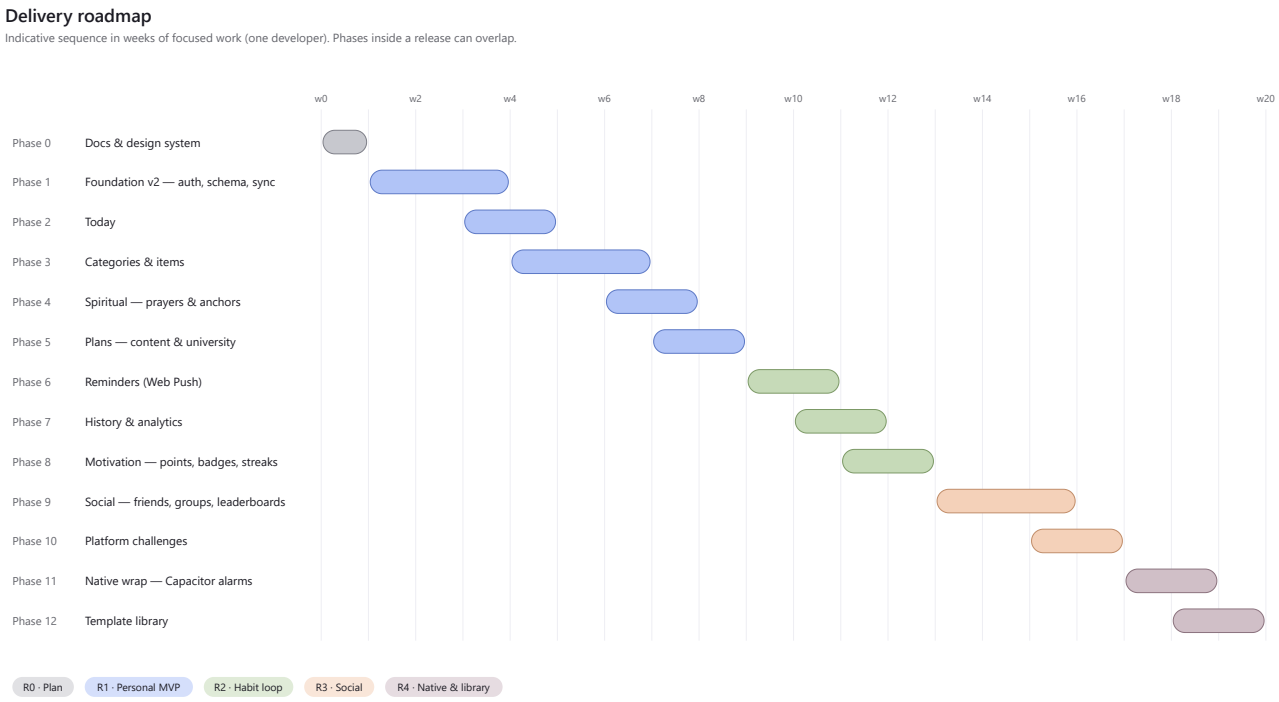
- No side color bars on cards. No invented hex values. No eyebrow labels on every section.
- No more than one accent per card. Status color goes on the tile or icon, never on the whole row.
- No hidden gestures as the only way to act: long-press always has a visible "⋯" twin.
- No gradients or illustrations. Ref 4's florals are decoration we deliberately leave out, for "simple".

8. Phase 0 deliverable

- A static **UI kit page** (`/dev/ui-kit`, dev-only) showing every component above in light and dark at 430 px.
- The owner approves it before Phase 2 builds Today on it.

Life v2 — Development plan

Branch: `v2`, cut from `master` on 2026-10-07. Phase specs: `docs/phases/v2/`. Inputs: REQUIREMENTS_V2 · SYSTEM_DESIGN · PRD · DESIGN_SYSTEM_V2



Durations are indicative (one developer, focused weeks). Phases inside a release can overlap where dependencies allow.

Releases and gates

Release	Phases	Gate to start the next release
R0 Plan	0	Owner approves this pack and the UI kit
R1 Personal MVP	1–5	Owner uses R1 daily for one week. All MUSTs of phases 1–5 pass. Migrated history matches the old reports.
R2 Habit loop	6–8	Reminders verified on Android Chrome and an installed iPhone PWA. Points are idempotent under re-sync.
R3 Social	9–10	Two real accounts complete a group goal end to end. Privacy tests pass.
R4 Native and library	11–12	Store builds approved

Phases at a glance

#	Phase	Main deliverables	Depends on
0	Docs and design system	This pack, tokens, Poppins, UI kit page	—
1	Foundation v2	Supabase Auth, new schema, migration, generalised sync, IndexedDB store, BUG-002	0

#	Phase	Main deliverables	Depends on
2	Today	Status buckets, date strip, one-tap, skip / move / pending, progress rule, new shell and nav	1
3	Categories and items	Category and item CRUD, tracking kinds, priority, schedules, routines, item detail	2
4	Spiritual	Prayer group, adhan-js, anchors, day parts, insights	3
5	Plans	Plan steps, dependencies, presets, cadence	3
6	Reminders	Web Push, pg-boss, escalation, quiet hours	3 (4 for prayer triggers)
7	History and analytics	Progress page, summaries, backfill, missed rollover	3
8	Motivation	Points ledger, streaks and freezes, badges	7
9	Social	Friends, groups, group goals, leaderboards, shares	8
10	Platform challenges	Challenges, admin	8
11	Native wrap	Capacitor, alarms, share image	6
12	Template library	LIB-01…05	3

Working rules (unchanged from AGENT.md, restated for v2)

1. One phase at a time. Each phase ends with every MUST passing, `npm run test`, `lint`, and `typecheck` green, and a browser check at 430 px in light and dark.
2. Decisions go in `DECISION.md`, bugs in `PROJECT.md`, status in `PROGRESS.md`.
3. Domain logic lives in `packages/shared` as pure, tested functions. UI components stay thin.
4. Every new sync entity gets an ownership test in `apps/api`.
5. No UI ships without being built from the Phase 0 kit components.

What happens to the old code

Old	When it goes
`habit_definitions`, `habit_logs` tables	Read-only after the Phase 1 migration; dropped at the start of R2
`today-dashboard.tsx`, `prayer-tracker.tsx`, `health/*`, `goals/content` page, old `globals.css` themes, `ThemePicker`	Deleted in Phase 2, when the new shell replaces them
`computeMonthlyPrayerReport`, `summarize*`, `ProgressRing`, `WeeklyBars`	Kept and adapted in Phases 3, 4, and 7
`ensureDefaultUser`, `SYNC_API_KEY`	Removed in Phase 1
`docs/phases/phase-06…14`	Moved to `docs/phases/archive/` (superseded)

v2 Phase 0 — Docs and design system

Release: R0 · **Requirements:** UI-01 · **Depends on:** —

Goal

Agree on what v2 is and how it looks before writing feature code.

Scope

- This documentation pack:
- REQUIREMENTS_V2, PRD, SYSTEM_DESIGN, DEVELOPMENT_PLAN, DESIGN_SYSTEM_V2
- diagrams and the compiled PDF
- Tokens from DESIGN_SYSTEM_V2 in a new `globals.css` (light and dark). Poppins via `next/font`.
- UI kit components in `apps/web/src/components/ui/`:
- StatusTile, TaskRow
- QuantityRow (stepper with typed value)
- PrayerRow, CategoryCard
- SegmentedTabs, DateStrip, BottomNav
- Sheet, Ring, Bars
- A dev-only page `/dev/ui-kit` showing every component in every state.

Acceptance

- [x] The owner approves the docs pack, including the answers to the open questions in PRD §12. Done 2026-10-08; the answers are recorded in REQUIREMENTS_V2 §3a.
- [ ] The owner approves the UI kit at 430 px in light and dark. Built 2026-10-09 at `/dev/ui-kit` (dev only), awaiting review.
- [x] Every text/background pair in the kit passes AA contrast. Enforced by `lib/theme/tokens.test.ts`, which also keeps `globals.css` in sync with `tokens.ts`.
- [x] Component tests: the stepper accepts ±step and typed values; the ring renders 0 / 50 / 100%. See `components/kit/kit.test.tsx`.

v2 Phase 1 — Foundation v2

Release: R1 · **Requirements:** ACC-01…03, NFR-01…05, migration (REQUIREMENTS_V2 §6) · **Depends on:** 0

Goal

Multi-user accounts on the new data model, with the owner's existing history carried over.

Scope

- **Auth:**
- Supabase Auth: email and password, password reset, Google optional (PRD §12 Q4)
- Web sign-in, sign-up, and reset screens, with a route guard
- API verifies the JWT via JWKS
- Remove `ensureDefaultUser` and `SYNC_API_KEY`
- **Schema:** a Drizzle migration for the core tables in SYSTEM_DESIGN §4:
- `users`, `user_settings`, `devices`
- `categories`, `items`, `item_dependencies`, `cadence_rules`
- `occurrences`, `reminders`, `templates`
- provenance columns
- **Shared Zod:** `Tracking`, `Schedule`, `Anchor`, `Trigger`, and the entity schemas in `packages/shared`.
- **Sync:**
- entity-agnostic outbox and pull
- ownership checks on every referenced id (fixes BUG-002)
- last-write-wins by `updated_at`
- soft deletes
- **Local store:**
- one SQLite DB per user, persisted to IndexedDB
- wiped on sign-out
- a cached session opens offline
- **Migration script:** old `habit_definitions` / `habit_logs` → categories, items, and occurrences. The owner's first sign-up claims the old Owner data.
- **Ops:** fix B-001. Either make `db:migrate:manual` the documented default on Windows, or patch drizzle-kit.

Acceptance

- [x] Two accounts on one device see completely separate data. A crafted push with a foreign id is rejected.
- Unit coverage: `sync/sync.test.ts` and `lib/v2/v2-store.test.ts`.
- Live check on 2026-10-09 against Supabase with two real accounts, 11/11 passed:
    - a foreign reference is rejected (422)

    - reusing another account's id is rejected (409)
    - pushing rows that claim another user is rejected (403)
    - stale edits are ignored
- [~] Sign-out wipes local data; a previously signed-in session opens offline.
- Wipe: browser E2E passed 9/9. It found and fixed a bug where an open store wrote itself back after the wipe.
- Session offline: the Supabase session is read from local storage with no network call.
- Full offline app launch still needs the production service worker, which is disabled in dev. Check this in Phase 2.
- [x] A new account starts with zero categories, and gets default settings (Karachi, Hanafi, friends visibility). Verified live.
- [~] After migration, the monthly prayer report and the steps, water, and sleep summaries give the same numbers on the new tables.
- Fixture test: prayer done/missed totals and step totals match v1. Sleep minutes and bed/wake times are carried over.
- Still open: running it on the owner's real data. That waits for the owner's sign-up, then `npm run db:migrate:v1 -w @life/api -- --email … [--apply]`.
- [x] The local DB survives more than 5 MB of data (IndexedDB). Covered by `lib/v2/v2-store.test.ts`.

v2 Phase 2 — Today

Release: R1 · **Requirements:** TOD-01…05, CAT-04, ITM-04, TRK-01, TRK-05, SCH-01, SCH-05, SCH-06, FLX-01…05, UI-02, UI-04 · **Depends on:** 1

Goal

Home opens on today's **To Do / In Progress / Done**. Days are flexible and can be planned ahead.

Scope

- **New app shell.** Bottom nav: Today · Categories · + · Progress · Profile. Delete the old dashboard, prayer, health, and goals screens and the old themes.
- **Shared logic** in `packages/shared`, with tests:
- `ScheduleExpander`, covering once and daily schedules only in this phase
- `ProgressRule`
- **Today screen:**
- header ring
- DateStrip covering past and future days
- StatusTiles that filter the list
- three status lists, plus a collapsed "Not today" section
- **Interactions:**
- one-tap check
- quantity stepper
- Start, which moves a task to In Progress
- long-press or ⋯ menu: Skip, Move to tomorrow, Move to date, Pending, Edit
- **Quick add (+):**
- title
- date chip: Today, Tomorrow, or a picked date; no date means Pending
- optional category and priority
- **Pending list,** reachable from Today.
- **Dark mode** using the derived tokens.

Acceptance

- [ ] Today puts items in the correct buckets. A quantity at 4,000 of 10,000 shows as In Progress.
- [ ] Skip, Move, and Pending each remove the item from today's required count, and the ring can still reach 100%.
- [ ] A day with nothing required shows 100% and "Nothing planned — add something?"
- [ ] At 21:00, a task added for tomorrow appears on tomorrow's list immediately, including when offline.
- [ ] Browser check passes at 430 px in light and dark. Tests cover the edge cases of the progress rule.

v2 Phase 3 — Categories and items

Release: R1 · **Requirements:** CAT-01…03, CAT-05, ITM-01…03, ITM-05…07, TRK-02…04, SCH-02, HIS-01 (basic) · **Depends on:** 2

Goal

Users build their own sections and decide, for each item, what it tracks, how important it is, and when it happens.

Scope

- **Categories page:**
- category cards showing today's ring and counts
- expand a category inline
- reorder by drag or with move buttons
- create, rename, set icon and color, archive, delete
- **Category detail:** items grouped by type; reorder; pin to top.
- **Full item editor** ("More options" from +):
- type
- tracking: check; quantity (unit, target, step); duration; checklist
- priority, plus an optional flag
- schedule: once, daily, weekdays, every N days, or monthly, with active from/until dates
- notes and extra fields
- **Sub-items:** routines, e.g. "Gym — Push day" with its exercises as children or as a checklist.
- **Item detail:**
- today's control
- schedule summary
- basic history: month heatmap, 7-day bars, completion rate, current streak
- **Templates and starter packs:**
- `GET /v1/templates` and `apply`
- onboarding picker: Spiritual, Physical, Health, Mental, Content creation, University, Work, or "Start empty"
- provenance recorded on applied items

Acceptance

- [ ] Create "Physical" with Steps (quantity, target 10,000, step 500). Log +500 twice, then type 10000. Works from both Today and the detail page.
- [ ] Create "Gym — Push day" with 5 exercises. Ticking all children completes the parent.
- [ ] A habit scheduled for weekdays [Mon, Wed, Fri] appears only on those days.
- [ ] Archiving a category hides it everywhere and keeps its history.

- [ ] A new user picks two starter packs, and only those appear.

v2 Phase 4 — Spiritual: prayers and anchors

Release: R1 · **Requirements:** PRY-01…07, SCH-03, SCH-04, TOD-06, TOD-07 · **Depends on:** 3

Goal

A customisable prayer block, one-tap marking in any order, prayer-time anchors, and honest insights.

Scope

- **Settings:**
- location, from device geolocation or a city search
- calculation method and madhab
- manual override per prayer
- `PrayerTimes` (adhan-js) in `packages/shared`. The client uses it now; the server reuses it later.
- **"Daily prayers" group item:**
- five default children
- rename, add, remove, and reorder prayers
- placement on Today and in any category
- **PrayerRow and ring:**
- one tap marks a prayer as prayed
- long-press offers Missed, Qada, and Undo
- no order is enforced
- **Anchors in the item editor:** before, at, or after a prayer, with an offset. Can be combined with weekdays, e.g. Kahf on Friday after Zuhr.
- **Day parts:** user-defined, optionally prayer-based. Today gets a toggle to group by day part.
- **Prayer insights:**
- totals per week and month
- rate per prayer
- most missed and most consistent prayer
- suggestion text
- reuses `computeMonthlyPrayerReport`

Acceptance

- [ ] Mark Asr while Fajr is unmarked, then mark Fajr as qada. Both count.
- [ ] "Surah Kahf · Friday · after Zuhr" appears only on Fridays, highlighted between Zuhr and Asr. It does not appear on other days.
- [ ] Renaming Zuhr → Dhuhr and adding Tahajjud updates the ring, insights, and history.
- [ ] Insights for a fixture month match hand-calculated totals, including the most-missed prayer.
- [ ] Karachi prayer times on a fixture date match a reference source within 1 minute.

v2 Phase 5 — Plans: content, university, cadence

Release: R1 · **Requirements:** PLN-01…06 · **Depends on:** 3

Goal

Multi-step, dated plans for things like restaurant collabs and university work, with optional cadence rules.

Scope

- **Plan detail:**
- steps on a timeline, with dates
- venue, contact, and notes
- a lock icon on dependent steps, with the reason
- **Editing `item_dependencies`:** "this step waits for…".
- **Presets:**
- **Content collab:** Shoot, Edit, Payment cleared, Post. Post depends on Payment.
- **University:** Assignment, Quiz, Presentation, Teacher meeting, FYP task or meeting, plus a course field.
- **Cadence rules** per category or plan:
- gap days between steps
- allowed weekdays per step kind
- `CadenceChecker` shows warnings and the next valid dates in the date picker.
- **Behaviour:**
- Steps appear on Today on their dates.
- The last step done completes the plan.

Acceptance

- [ ] "Collab — Café X" with shoot Tue, edit Wed, post Fri. Post stays locked until Payment cleared is ticked.
- [ ] Rule "post weekdays Tue/Fri, gap ≥ 1 day": picking Thursday shows a warning and suggests Friday. Saving anyway is still allowed.
- [ ] Quiz on 12 Oct 10:00 for course "DSA" appears on Today on 12 Oct.
- [ ] A content category with no active plans shows "Nothing upcoming" and no warning.

v2 Phase 6 — Reminders

Release: R2 · **Requirements:** REM-01…06 · **Depends on:** 3 (and 4 for prayer-relative triggers)

Goal

Reminders that match importance: critical items keep nudging until done.

Scope

- **Push setup:**
- VAPID keys
- `push_subscriptions` endpoints
- Service worker handles `push` and `notificationclick`, with **Done** and **Snooze** actions
- **Scheduling:** pg-boss, in the API process or a separate worker. Jobs: `reminders.tick` (every minute) and `reminders.escalate`.
- **Reminder editor on items:**
- at a set time
- N minutes before due
- relative to a prayer
- **Priority defaults:** per REM-02, editable.
- **Settings:**
- quiet hours
- a global switch
- per-category mute
- **iOS:** a guide to install to the Home Screen, shown when the user turns on notifications in Safari.
- **Optional nudges:** an evening "Plan tomorrow" nudge and a morning summary.

Acceptance

- [ ] A medium reminder fires once at the set time, on Android Chrome and on an installed iPhone PWA.
- [ ] A critical reminder repeats every 30 min. It stops as soon as the item is done, including when Done is tapped on the notification.
- [ ] Nothing fires during quiet hours. Muting a category silences its items.
- [ ] A prayer-relative reminder (10 min after Asr) fires at the right local time across a DST change (unit test).

v2 Phase 7 — History and analytics

Release: R2 · **Requirements:** HIS-01…04, FLX-06 · **Depends on:** 3

Goal

Clear week and month views of how things are going, per item, per category, and overall.

Scope

- **Progress page:**
- overview of the day % over the week and the month
- per category: completion, skipped, and moved counts
- "doing well / needs attention"
- **Item history, completed:**
- log list with values
- best streak
- skip and move counts
- **Day rollover:** unresolved required occurrences count as **missed** in stats and streaks. They never carry over to the next day.
- **Backfill:** past days can be edited from the date strip or the item history. Late edits earn no points after 48 h; Phase 8 enforces this.
- **Stats code:** the old health and sleep stats (`summarizeCounter`, `summarizeSleep`, …) now read from occurrences.

Acceptance

- [ ] Week and month summaries match a seeded fixture.
- [ ] A required item left open yesterday shows as missed in history and is not in today's list.
- [ ] Editing last Tuesday's steps updates that week's chart.

v2 Phase 8 — Motivation: points, streaks, badges

Release: R2 · **Requirements:** REW-01…04 · **Depends on:** 7

Goal

Rewards that encourage without punishing.

Scope

- **Server tables:** `points_ledger`, `streaks`, `badge_definitions`, `user_badges`.
- **`RewardsEngine`:**
- `evaluate`, triggered from sync when an occurrence is newly done
- `rollStreaks`, run at each user's local midnight
- `checkBadges`
- **Points:**
- awarded by priority
- a bonus for a perfect day
- the 48 h backfill rule applies
- **Streak freezes:**
- earn 1 per 7-day streak, hold at most 2
- used automatically when a day is missed
- gentle copy when a streak restarts
- **Badge set v1:**
- 3, 7, 30, and 100-day streaks
- all prayers for a week
- first plan finished
- perfect week
- planner: planned tomorrow 7 nights in a row
- **Progress page:** points, level, a badges grid, and streaks.
- **Celebration:** a small animation on completion, a perfect day, or a new badge. Respects reduced motion.

Acceptance

- [ ] Re-syncing the same completions twice leaves the points total unchanged (idempotency test).
- [ ] A 7-day streak earns a freeze. Missing day 8 uses it and the streak continues.
- [ ] A completion backfilled 3 days late earns 0 points but still counts in history.

v2 Phase 9 — Social: friends, groups, leaderboards, sharing

Release: R3 · **Requirements:** SOC-01…05, SHR-01 · **Depends on:** 8

Goal

Accountability with friends, without giving up privacy.

Scope

- **Friends:** invite by link or email; accept, remove, block.
- **Groups:**
- create a group
- invite registered friends, or share a link that goes through sign-up and then join
- member roles
- **Group goals:**
- title, tracking, period, reward
- each member gets a linked item on their own Today (`items.group_goal_id`)
- progress is aggregated in `group_goal_progress`
- **Leaderboards:**
- weekly and monthly snapshots
- individual rankings among friends and within a group
- group vs group, ranked by average points per active member
- rewards for the top 3
- **Shares:**
- send a streak, badge, or achievement to a friend or a group
- kudos reaction
- push notification to the recipient
- **Privacy tests:** no endpoint returns another user's item titles unless they were shared.

Acceptance

- [ ] Two real accounts: A invites B, B signs up via the link and joins, and both complete "70k steps this week". Both earn points and the group's points rise.
- [ ] The leaderboard shows only names, avatars, and points.
- [ ] Sharing a 21-day Fajr streak notifies the friend, and their kudos shows on the sharer's side.

v2 Phase 10 — Platform challenges

Release: R3 · **Requirements:** CHL-01, CHL-02 · **Depends on:** 8

Goal

The platform sets weekly and monthly goals. Users who finish in time earn rewards.

Scope

- **Server tables:** `challenges` and `challenge_entries`.
- Each challenge has a rule, stored as JSON and evaluated by `RewardsEngine`.
- Example rule: `{metric: 'prayers_done', gte: 30, period: 'week'}`.
- **User side:** a challenge list on Progress, a join button, and a progress bar.
- **Admin:**
- a minimal `/admin/challenges` screen, gated by the admin claim
- admins can create, schedule, and close challenges
- the `challenges.close` job awards points and badges

Acceptance

- [ ] A weekly challenge completed before the deadline awards points and a badge exactly once.
- [ ] Completing it after `ends_at` awards nothing.
- [ ] A non-admin cannot reach the admin routes (API test).

v2 Phase 11 — Native wrap (Capacitor)

Release: R4 · **Requirements:** REM-05 (alarms), SHR-02 · **Depends on:** 6

Goal

Real alarm-style reminders and a presence in the app stores, built from the same web code.

Scope

- A Capacitor project wrapping the Next.js build. Whether to use a static export or a hosted URL is decided in this phase.
- `@capacitor/local-notifications`, with exact alarms on Android.
- Critical reminders are scheduled locally.
- The local schedule is mirrored with the server so a reminder doesn't fire twice.
- A share sheet for streak and achievement images, generated client-side from the Ring and Bars components.
- Store listing assets, a privacy policy, and a TestFlight / internal testing track.

Acceptance

- [ ] A critical reminder rings on Android with the app closed and repeats until Done.
- [ ] iOS local notifications fire on time with the app closed.
- [ ] Sharing a streak image to WhatsApp and Instagram works.

v2 Phase 12 — Public template library

Release: R4 · **Requirements:** LIB-01…05 (unchanged from Addendum A) · **Depends on:** 3

Goal

Users publish their own category templates. Others can adopt and star them.

Scope

The rules and open questions in the archived phase-14-template-library carry over unchanged. They now apply to categories and items instead of habits. Provenance columns have existed since Phase 1.

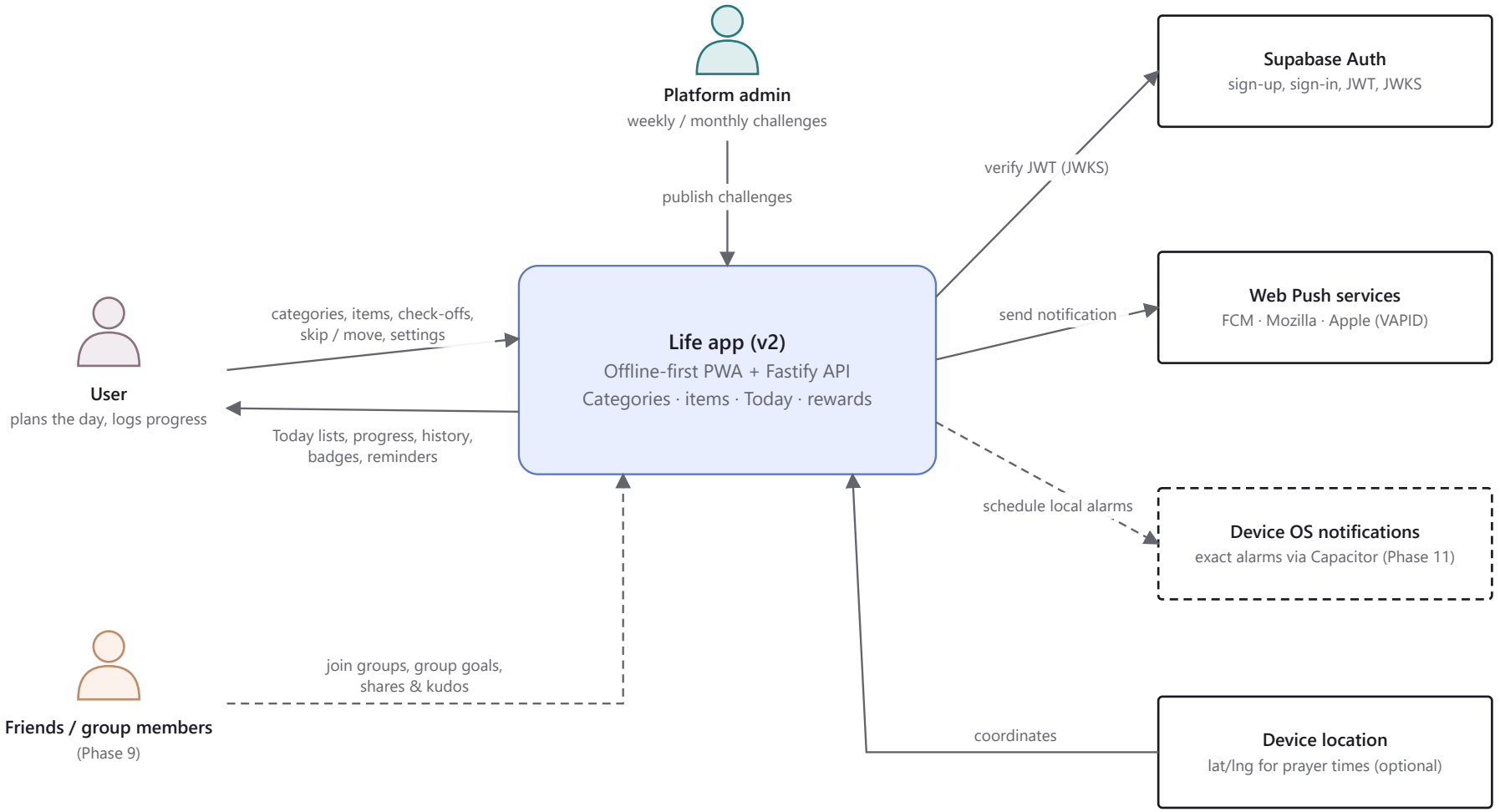
Acceptance

- [ ] Publishing fails if any item's provenance points to another author's template (LIB-04).
- [ ] Adopt, star, and sort-by-stars all work.

System context

System context

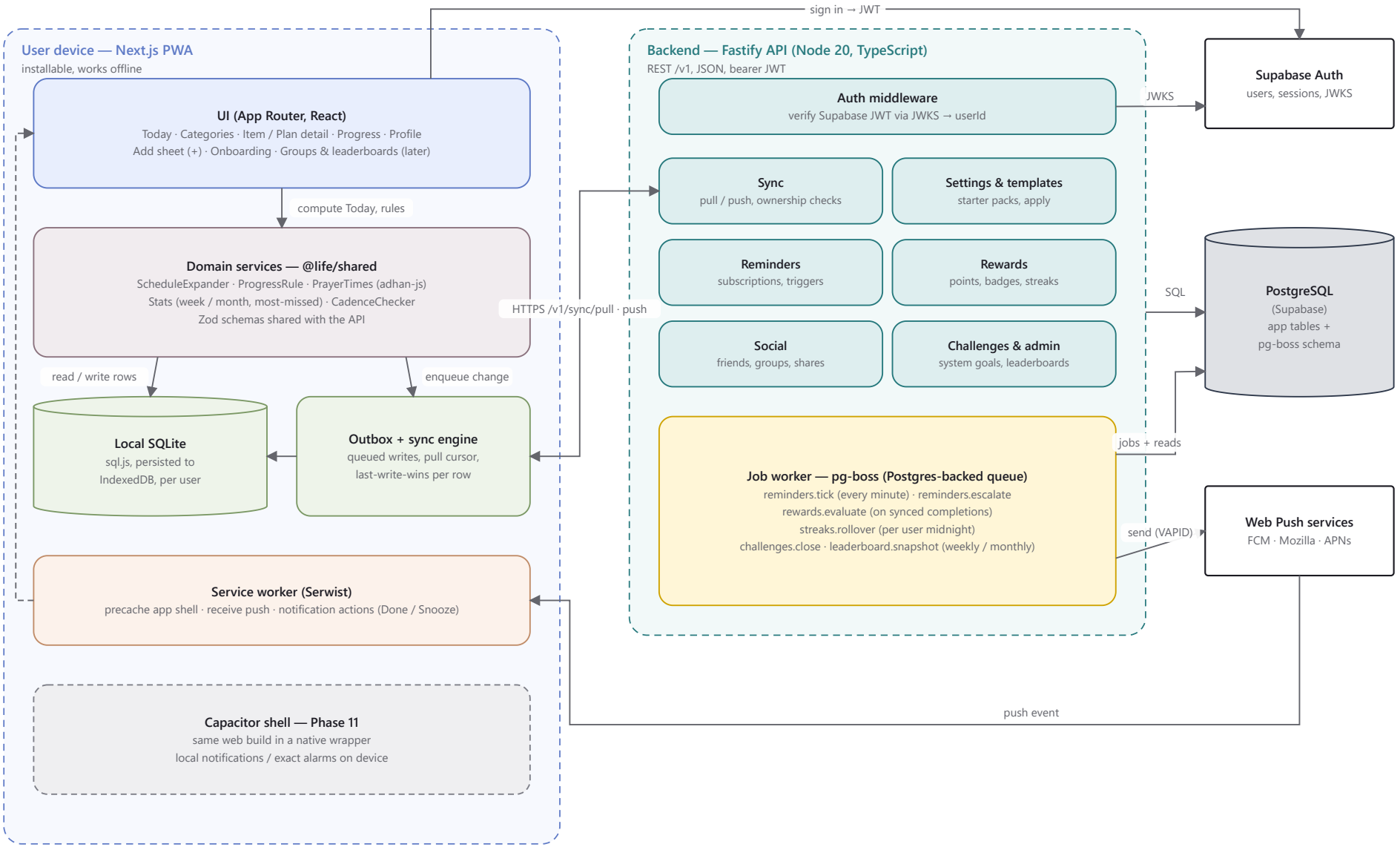
Who and what the Life app talks to. Solid = now (PWA), dashed = later phases.



System architecture

System architecture

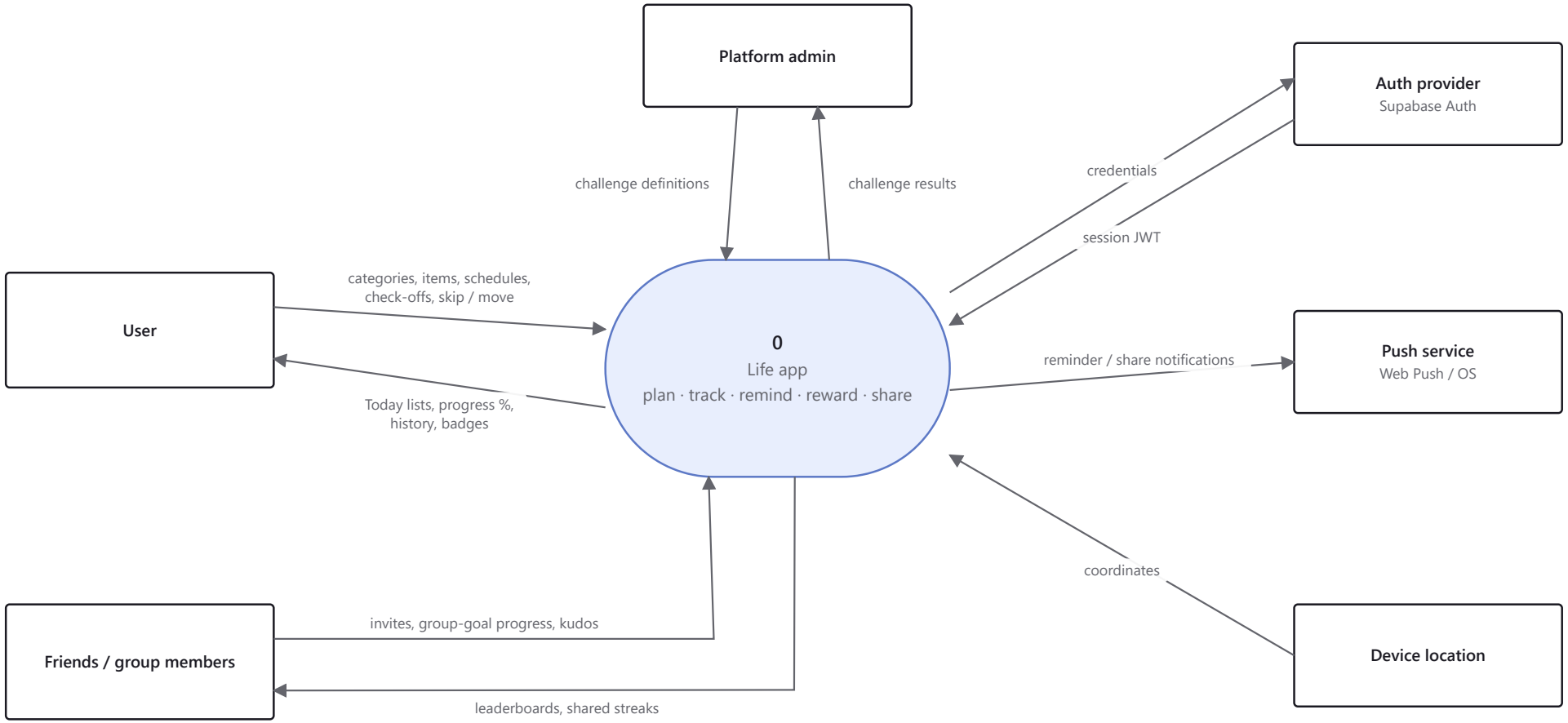
Containers and modules. Postgres is the source of truth; each device keeps a per-user SQLite cache and an outbox.



Data flow — level 0

Data flow — level 0

The whole system as one process. Rectangles = external entities, rounded = process.



Data flow — level 1

Data flow — level 1

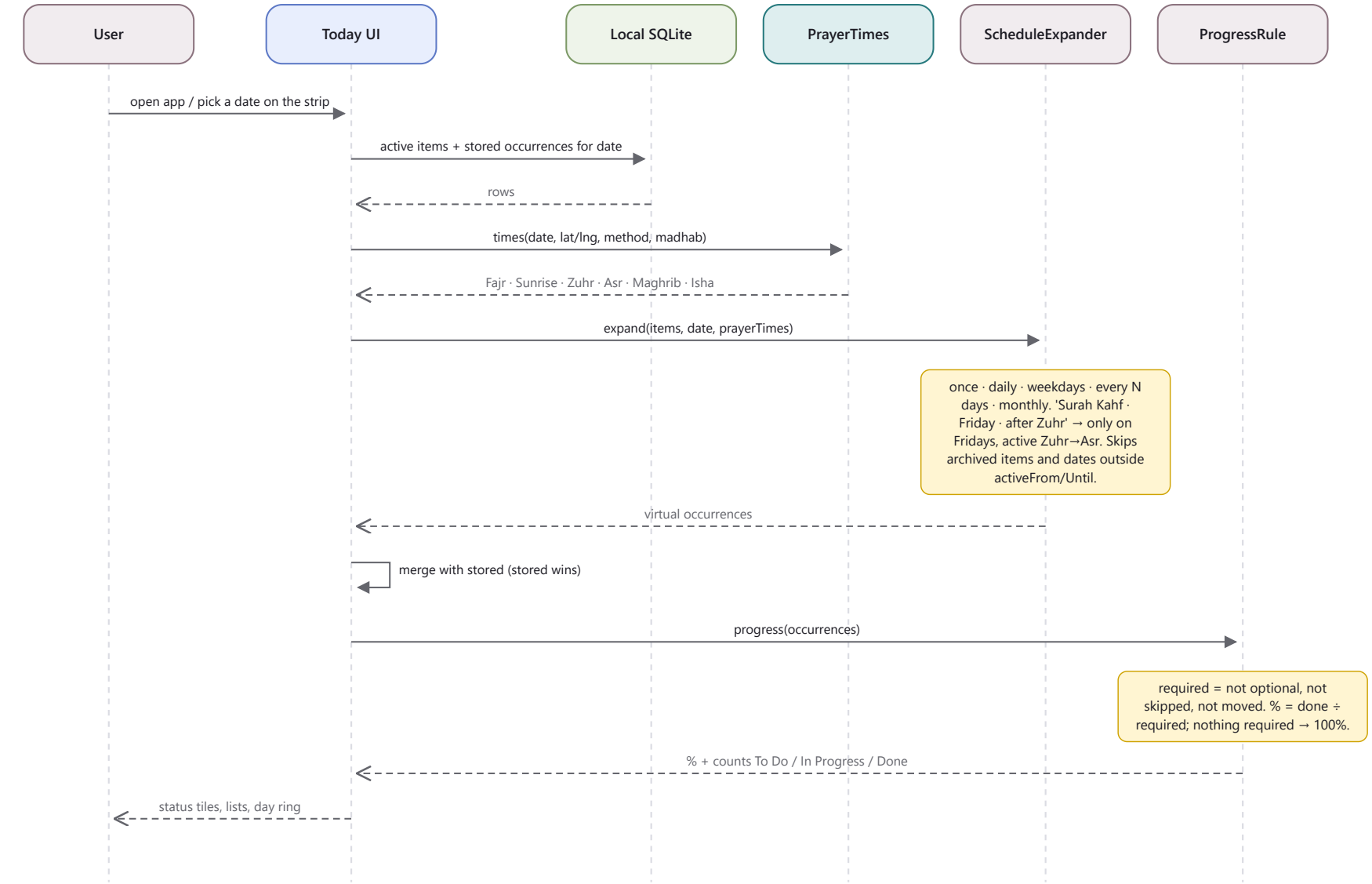
Processes 1–8, data stores D1–D7. Stores are logical: each exists on the server, and D1–D3 + D7 are cached on the device.



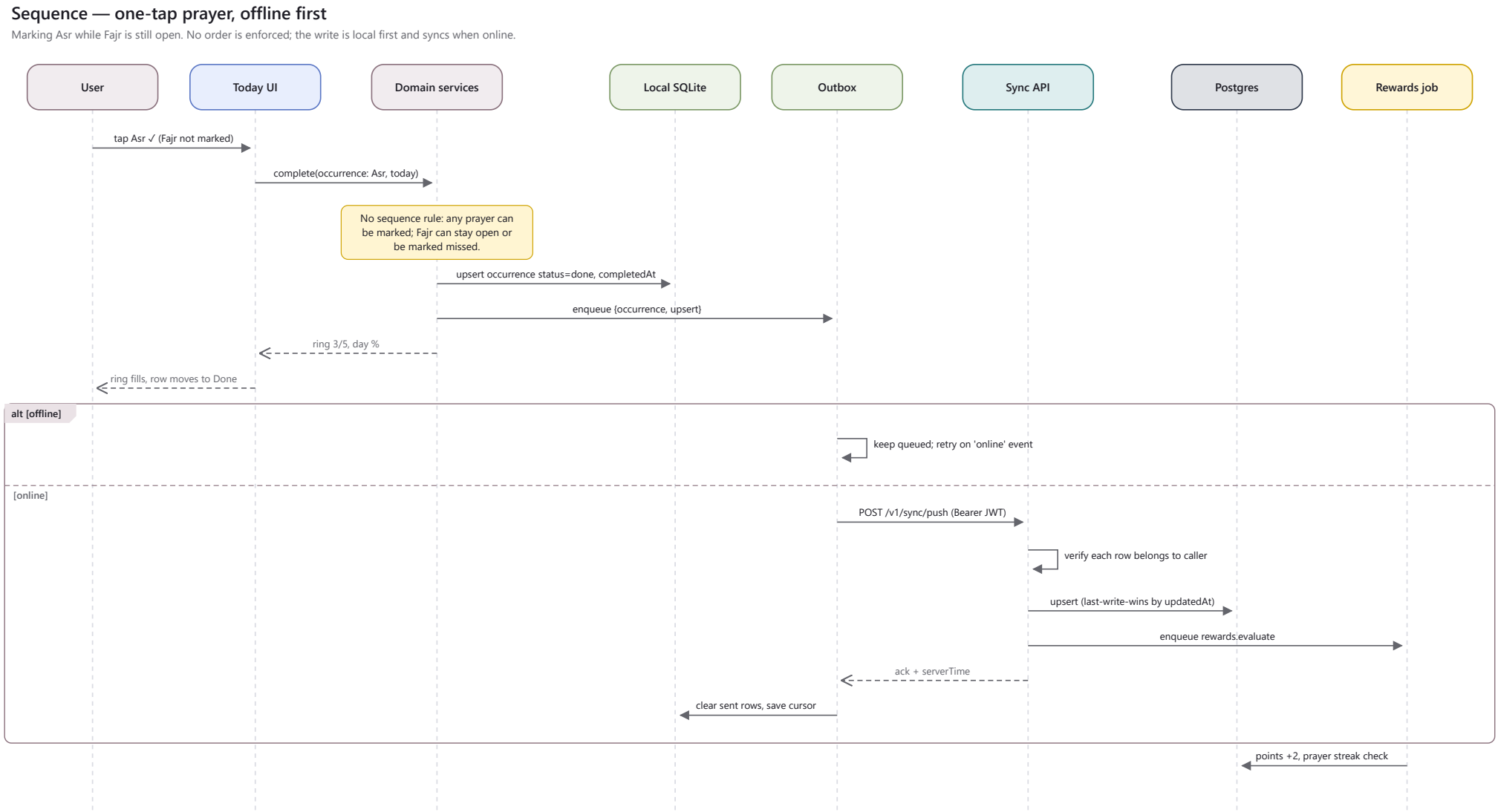
Sequence — building Today

Sequence — building Today

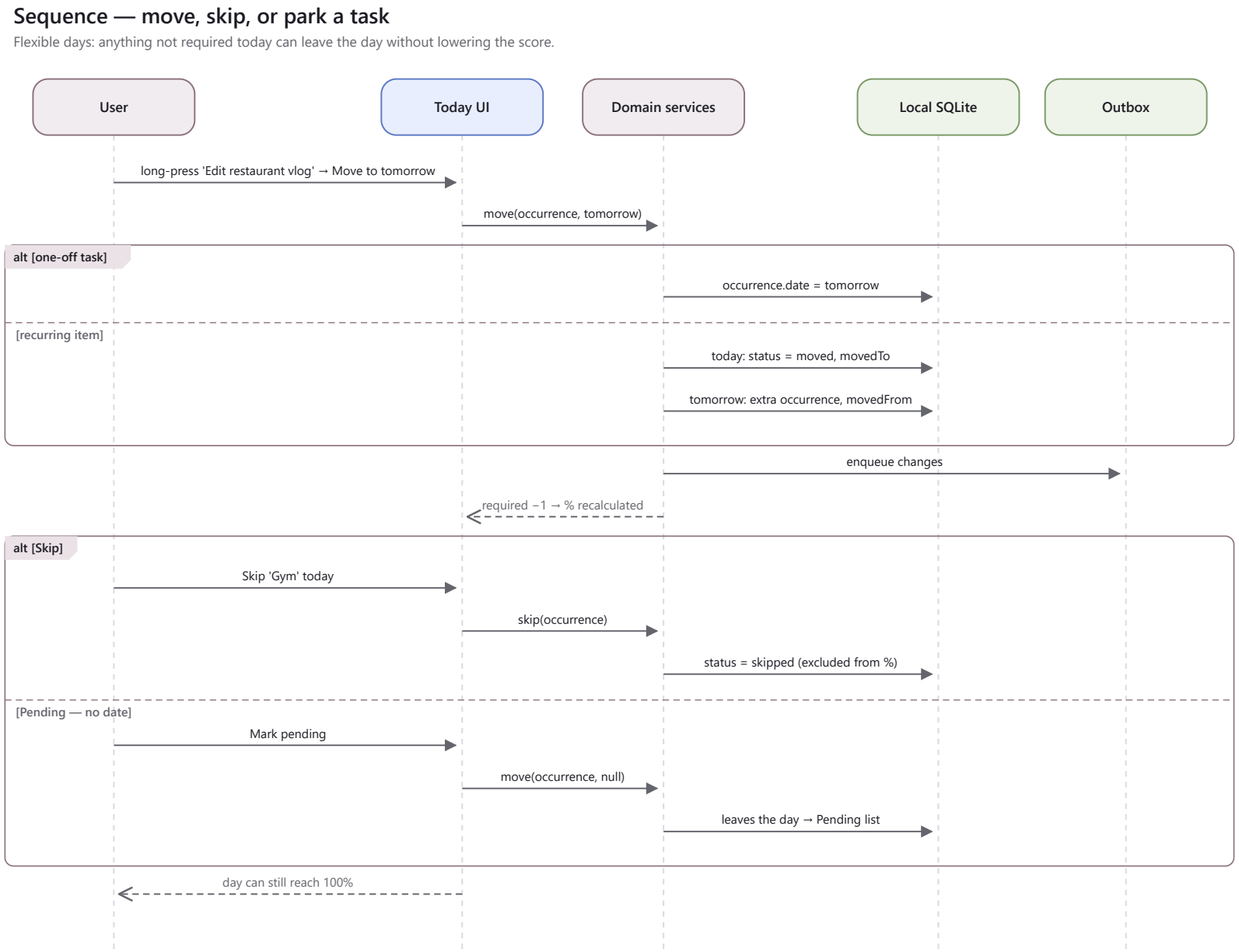
Occurrences are computed from schedules on the device and stored only once the user touches them.



Sequence — one-tap prayer, offline first



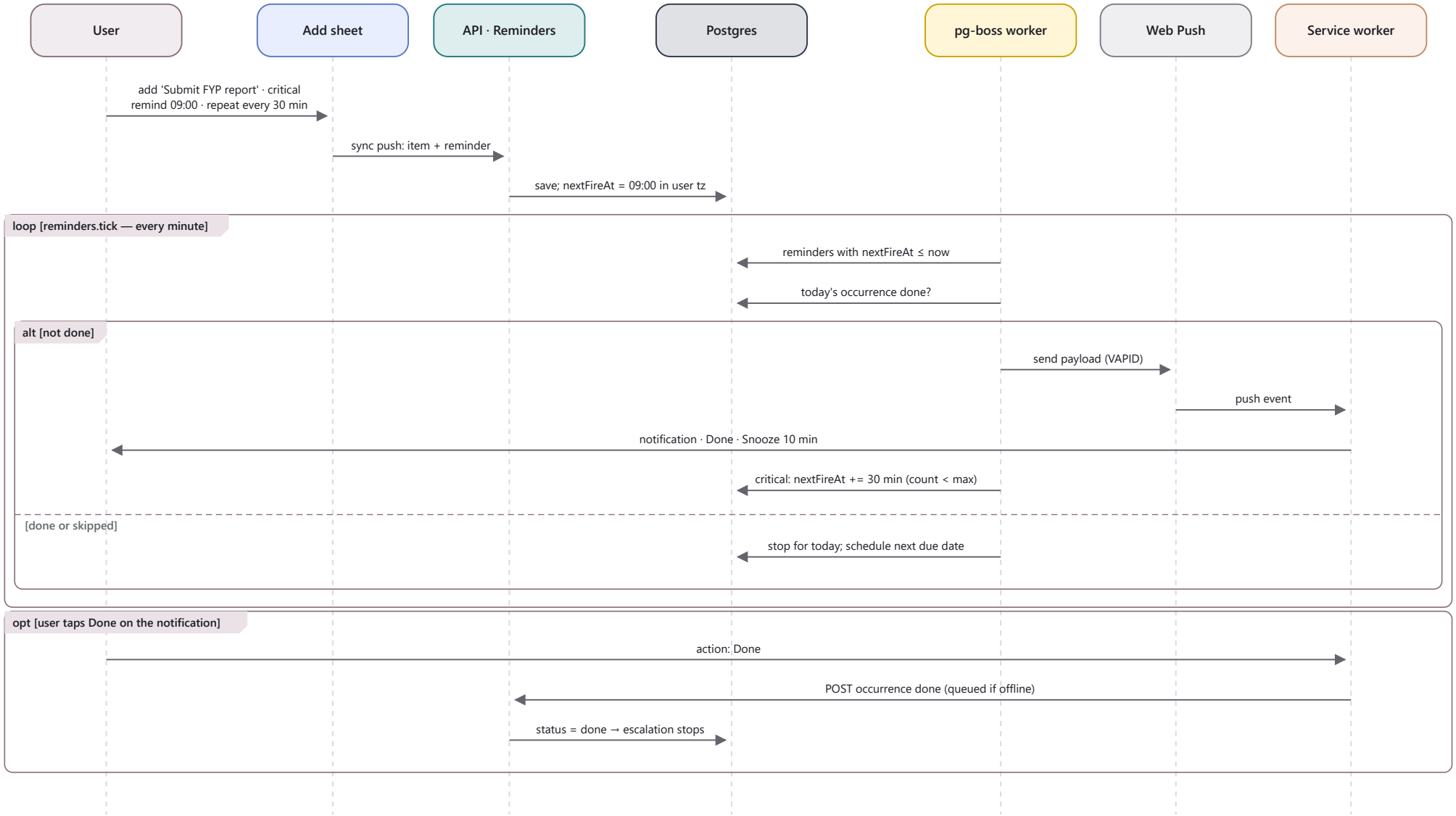
Sequence — move, skip, pending



Sequence — reminder escalation

Sequence — reminder with priority escalation

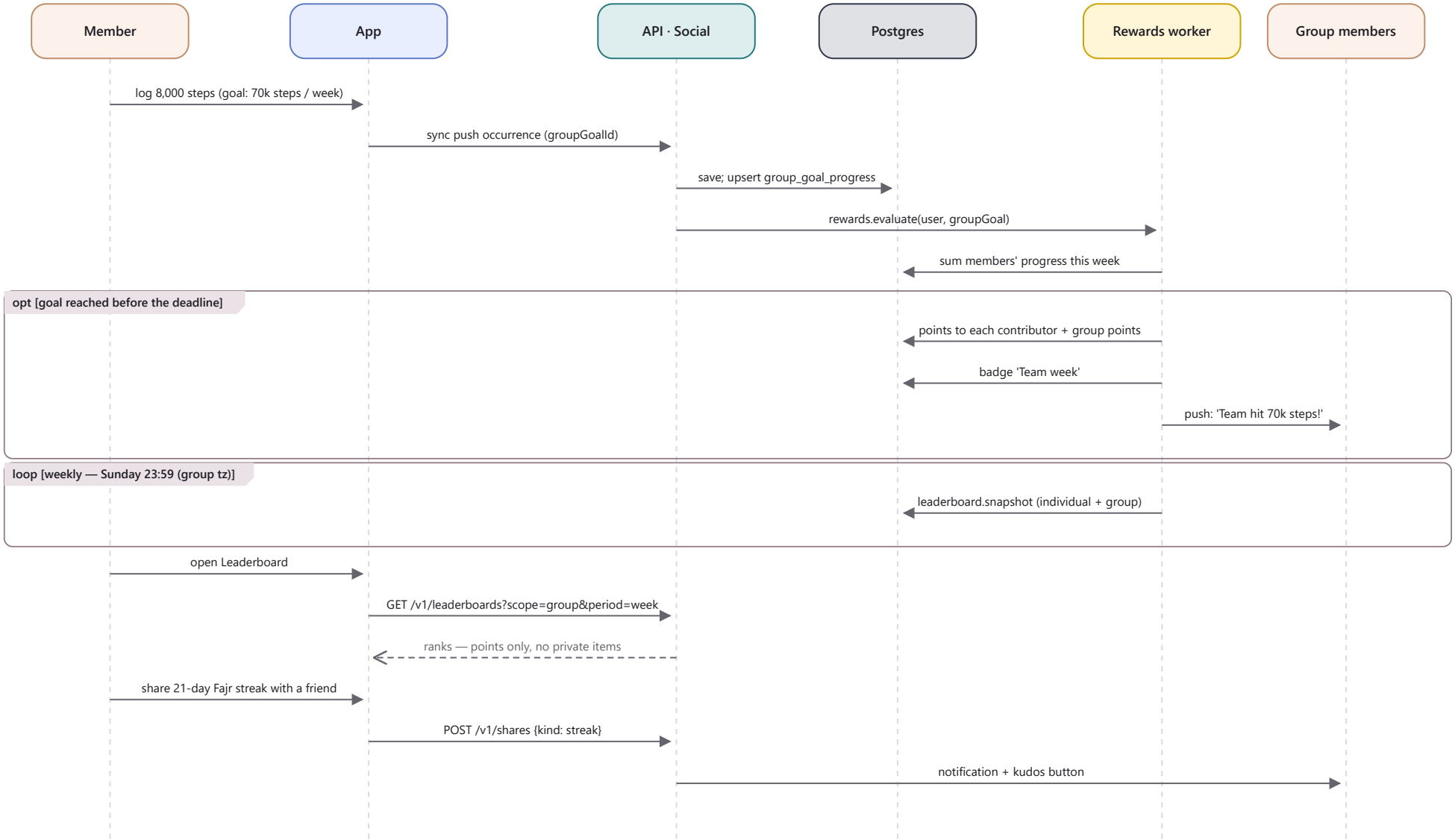
A critical item repeats its reminder until it is done (bounded by max repeats and quiet hours).



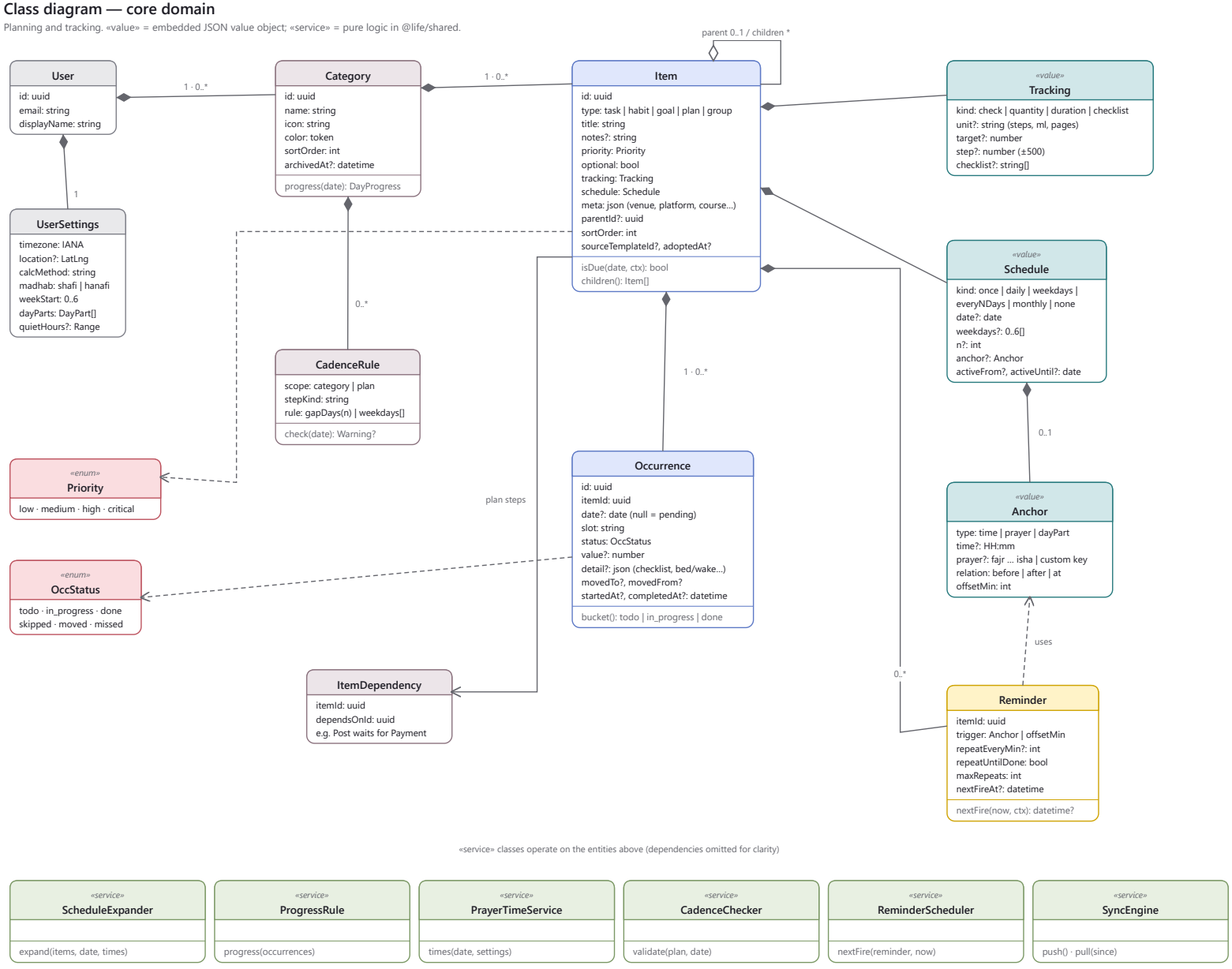
Sequence — group goal and leaderboard

Sequence — group goal, points and leaderboard (Phase 9)

Rewards are computed on the server only, from synced data, so they cannot be granted by a client.



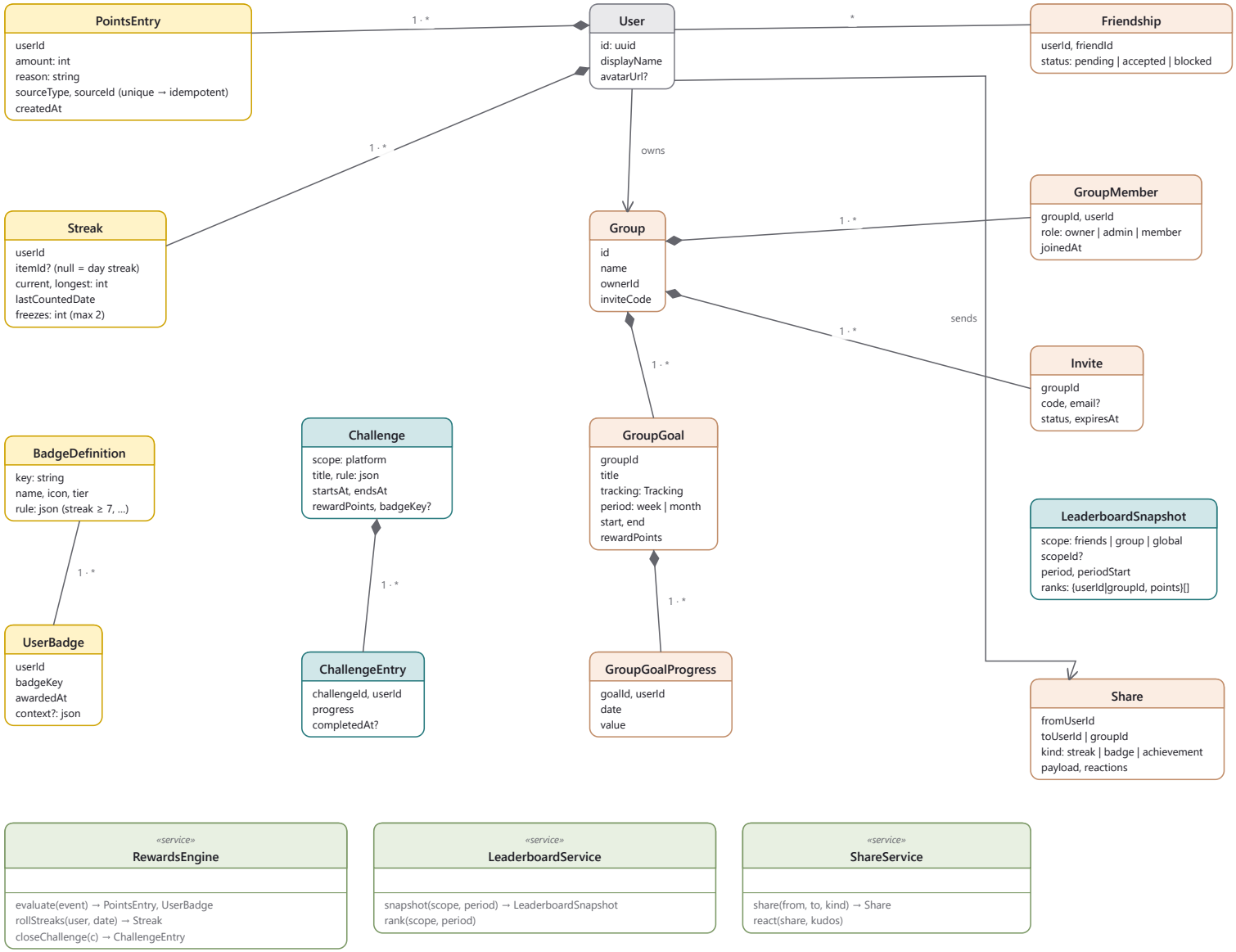
Class diagram — core



Class diagram — rewards and social

Class diagram — rewards & social

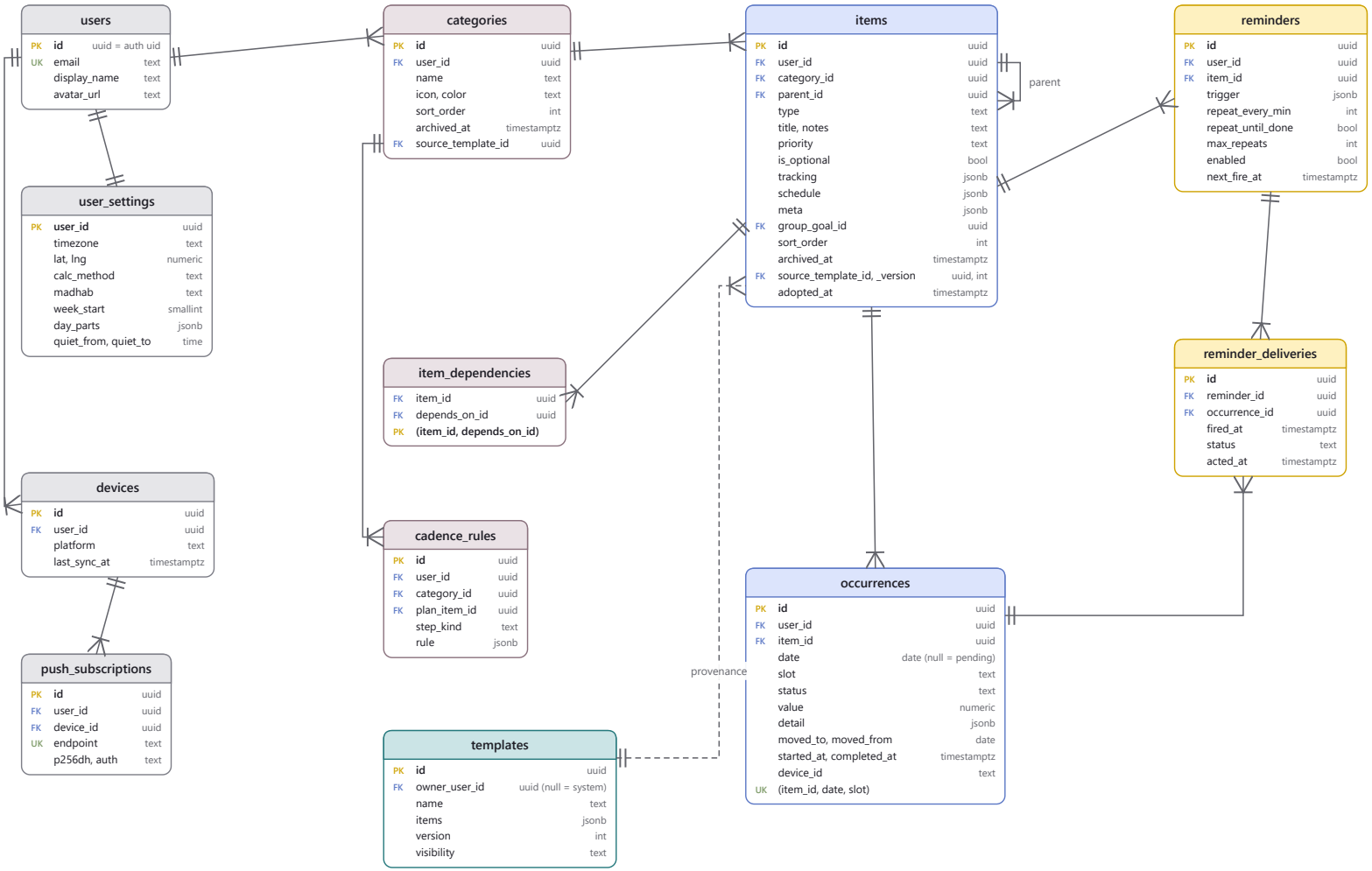
Motivation (Phase 8), social (Phase 9) and platform challenges (Phase 10). All written by server-side services only.



ERD — core

ERD — core tables

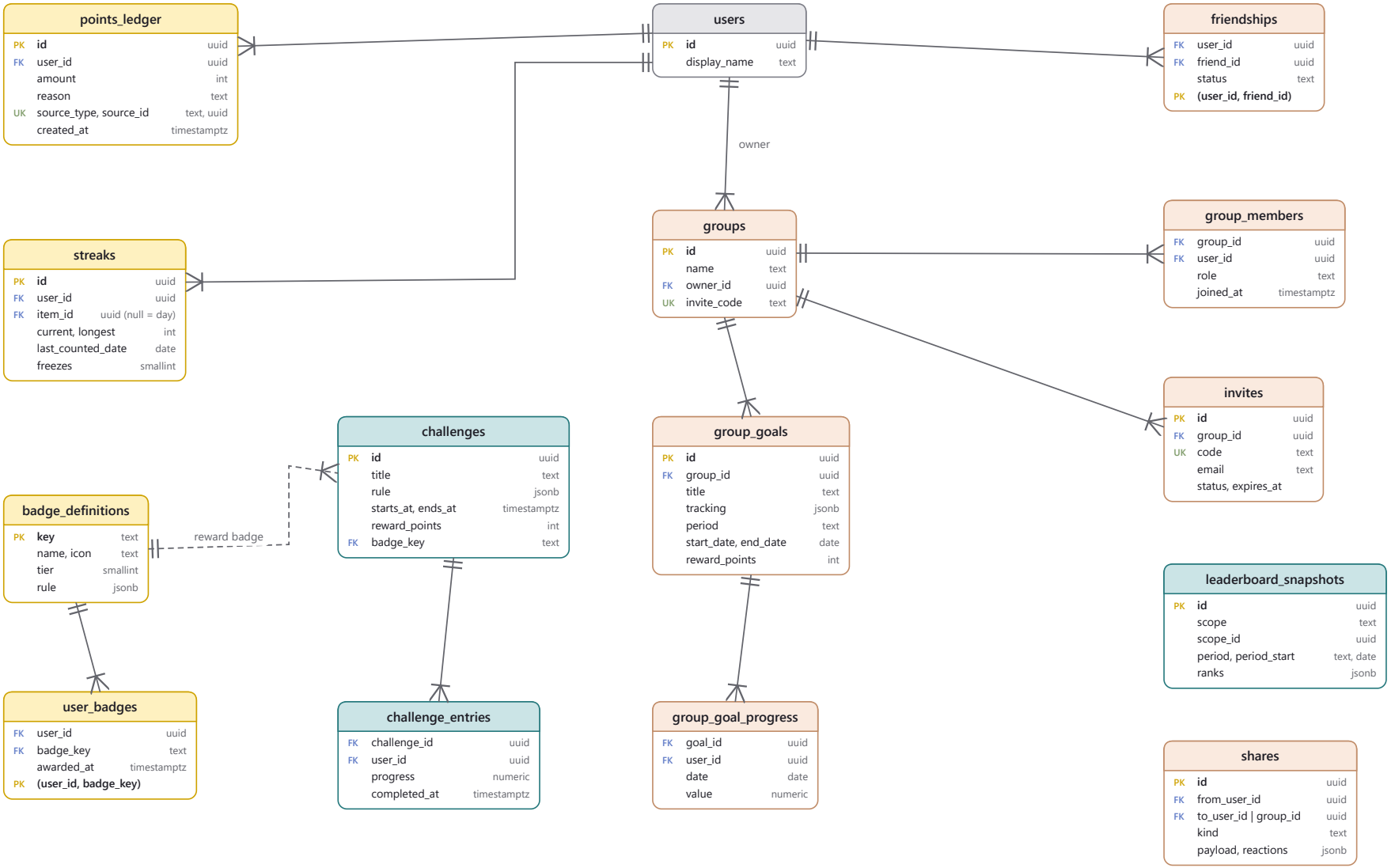
Postgres. Every table also has created_at, updated_at, deleted_at (soft delete, for sync). Local SQLite mirrors the user-owned tables.



ERD — rewards and social

ERD — rewards & social tables

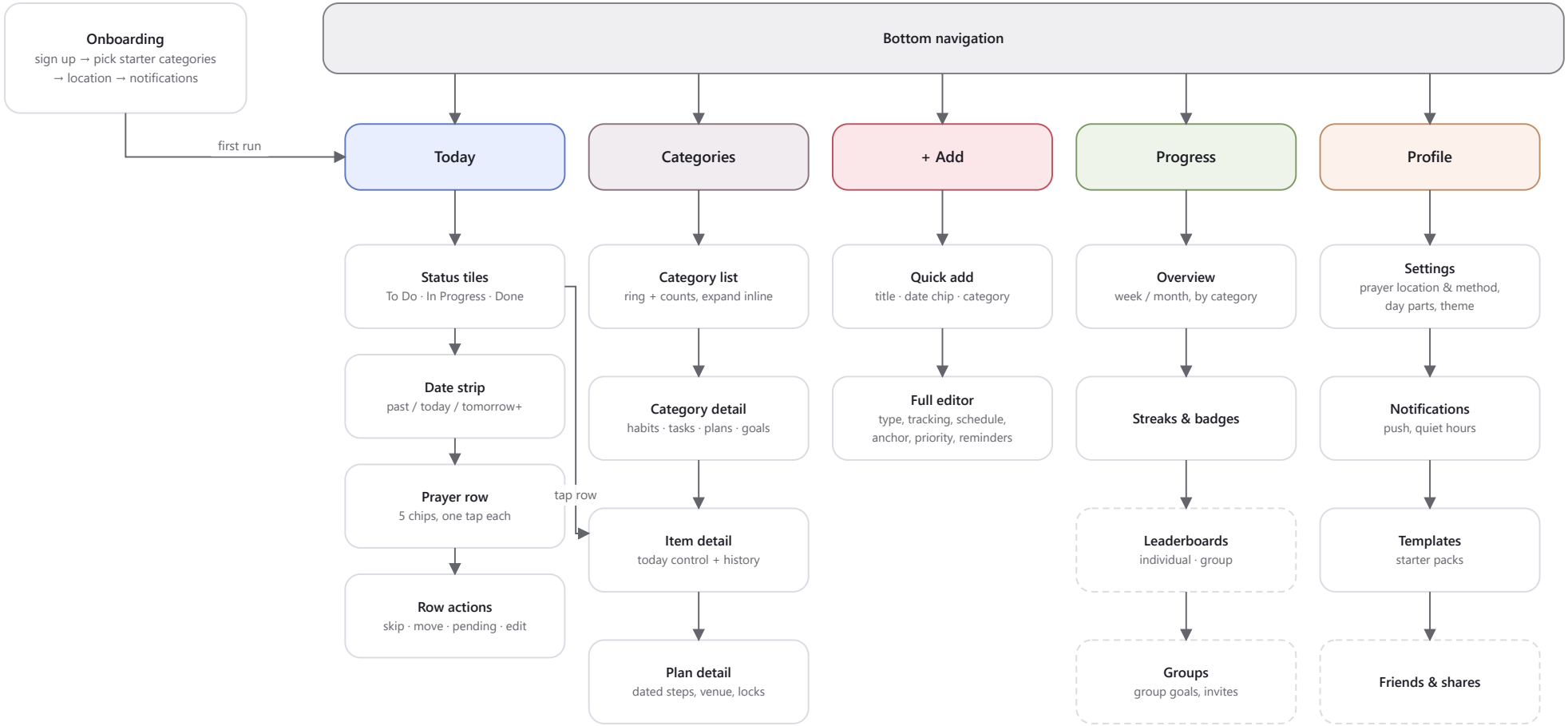
All rows written by the API / jobs, never by client sync. Leaderboards read points only; item details stay private.



Screen map

Screen map

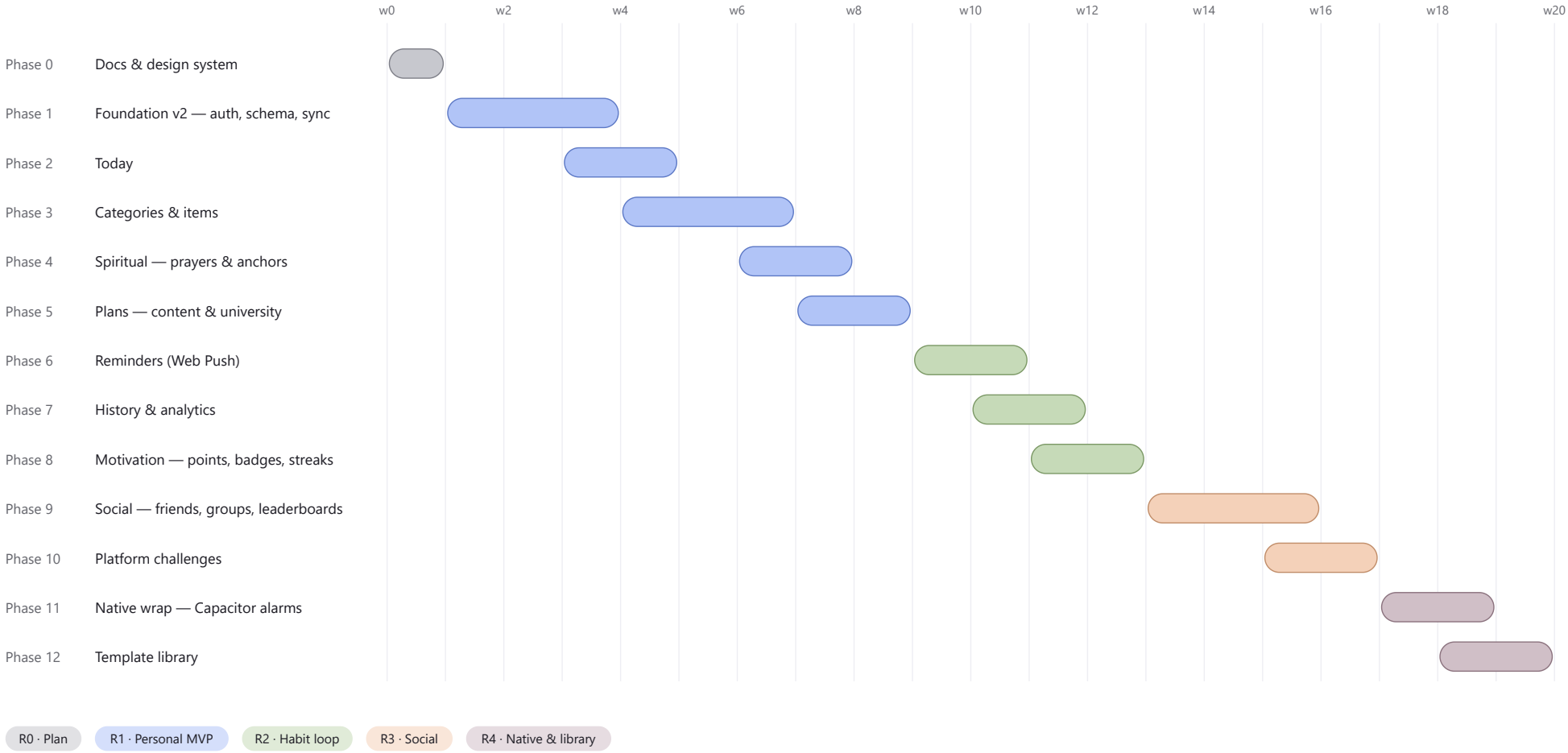
Bottom navigation: Today · Categories · + · Progress · Profile. Dashed = later phases.



Roadmap

Delivery roadmap

Indicative sequence in weeks of focused work (one developer). Phases inside a release can overlap.



Palette

v2 palette — sampled from UI Images/

Median of a 7×7 pixel patch at each element (scratch script). (d) = derived for contrast, not sampled.

Surfaces & ink

Background
#F4F4F9 · ref 2 screen

Surface
#FFFFFF · cards

Track / chip
#E5E5EF · ref 2

Border
#DBDCE3 · ref 2

Ink
#1B1A22 · ref 2 heading

Ink 2
#64656C · ref 2 body

Status

To Do
#E66473 · ref 3

In Progress
#EBAC80 · ref 2

Done
#97BB7D · ref 2

Skipped / neutral
#AA8B9A · ref 2

Primary
#7293F1 · ref 1

Primary text (d)
#4766D0 · AA on white

Category pastels

Mint
#D8E7E4 · ref 4

Lilac
#D7CCEA · ref 4

Peach
#E9CFC8 · ref 4

Lime
#E6ECCA · ref 4

Pink
#F6E4E2 · ref 4

Sky
#86CFE2 · ref 1